数字电路答案

第一章习题

1-1

（1）$22_{10} = 2*8^1 + 6*8^0 = 26_8$

$26_8 = \underbrace{2}_{010}\ \underbrace{6}_{110} = 10110_2$

$10110_2 = \underbrace{0001}_{1}\underbrace{0110}_{6} = 16_{16}$

（2） $108_{10} = 1*8^2 + 5*8^1 + 4*8^0 = 154_8$

$154_8 = \underbrace{1}_{001}\ \underbrace{5}_{101}\ \underbrace{4}_{100} = 1101100_2$

$1101100_2 = \underbrace{0110}_{6}\underbrace{1100}_{C} = 6C_{16}$

（3）$13.125_{10} = 1*8^1 + 5*8^0 + 1*8^{-1} = 15.1_8$

$15.1_8 = \underbrace{1}_{001}\ \underbrace{5}_{101}.\underbrace{1}_{001} = 1101.001_2$

$1101.001_2 = \underbrace{1101}_{D}.\underbrace{0010}_{2} = D.2_{16}$

（4）$131.625_{10} = 2*8^2 + 0*8^1 + 3*8^0 + 5*8^{-1} = 203.5_8$

$203.5_8 = \underbrace{2}_{010}\ \underbrace{0}_{000}\ \underbrace{3}_{011}.\underbrace{5}_{101} = 10000011.101_2$

$10000011.101_2 = \underbrace{1000}_{8}\underbrace{0011}_{3}.\underbrace{1010}_{A} = 83.A_{16}$

1-2

（1）$101101_2 = \underbrace{101}_{5}\underbrace{101}_{5} = 55_8$

$101101_2 = \underbrace{0010}_{2}\underbrace{1101}_{D} = 2D_{16}$

$55_8 = 5*8^1 + 5*8^0 = 45_{10}$

（2）$11100101_2 = \underbrace{011}_{3}\underbrace{100}_{4}\underbrace{101}_{5} = 345_8$

$11100101_2 = \underbrace{1110}_{E}\underbrace{0101}_{5} = E5_{16}$

$345_8 = 3*8^2 + 4*8^1 + 5*8^0 = 229_{10}$

（3）$101.0011_2 = \underbrace{101}_{5}.\underbrace{001}_{1}\underbrace{100}_{4} = 5.14_8$

$101.0011_2 = \underbrace{0101}_{5}.\underbrace{0011}_{3} = 5.3_{16}$

$5.14_8 = 5*8^0 + 1*8^{-1} + 4*8^{-2} = 5.1875_{10}$

（4）$100111.101_2 = \underbrace{100}_{4}\underbrace{111}_{7}.\underbrace{101}_{4} = 47.4_8$

$100111.101_2 = \underbrace{0010}_{2}\underbrace{0111}_{7}.\underbrace{1010}_{A} = 27.A_{16}$

$47.5_8 = 4*8^1 + 7*8^0 + 5*8^{-1} = 39.625_{10}$

1-3

（1）$16_8 = 1*8^1 + 6*8^0 = 14_{10}$

$16_8 = \underbrace{1}_{001}\ \underbrace{6}_{110} = 1110_2$

$1110_2 = \underbrace{1110}_{E} = E_{16}$

（2）$172_8 = 1*8^2 + 7*8^1 + 2*8^0 = 122_{10}$

$172_8 = \underbrace{1}_{001}\ \underbrace{7}_{111}\ \underbrace{2}_{010} = 1111010_2$

$1111010_2 = \underbrace{0111}_{7}\underbrace{1010}_{A} = 7A_{16}$

（3）$61.53_8 = 6*8^1 + 1*8^0 + 5*8^{-1} + 3*8^{-2} = 49.672_{10}$

$61.53_8 = \underbrace{6}_{110}\ \underbrace{1}_{001}.\underbrace{5}_{101}\ \underbrace{3}_{011} = 110001.101011_2$

$110001.101011_2 = \underbrace{0011}_{3}\underbrace{0001}_{1}.\underbrace{1010}_{A}\underbrace{1100}_{C} = 31.AC_{16}$

（4）$126.74_8 = 1*8^2 + 2*8^1 + 6*8^0 + 7*8^{-1} + 4*8^{-2} = 86.9375_{10}$

$126.74_8 = \underbrace{1}_{001}\ \underbrace{2}_{010}\ \underbrace{6}_{110}.\underbrace{7}_{111}\ \underbrace{4}_{100} = 1010110.1111_2$

$1010110.1111_2 = \underbrace{0101}_{5}\underbrace{0110}_{6}.\underbrace{1111}_{F} = 56.F_{16}$

1-4

（1）$2A_{16} = \underbrace{2}_{0010}\ \underbrace{A}_{1010} = 101010_2$

$101010_2 = \underbrace{101}_{5}\underbrace{010}_{2} = 52_8$

$52_8 = 5*8^1 + 2*8^0 = 42_{10}$

（2） $B2F_{16} = \underbrace{B}_{1011}\ \underbrace{2}_{0010}\ \underbrace{F}_{1111} = 101100101111_2$

$101100101111_2 = \underbrace{101}_{5}\underbrace{100}_{4}\underbrace{101}_{5}\underbrace{111}_{7} = 5457_8$

$5457_8 = 5*8^3 + 4*8^2 + 5*8^1 + 7*8^0 = 2863_{10}$

（3） $D3.E_{16} = \underbrace{D}_{1101}\ \underbrace{3}_{0011}\ .\ \underbrace{E}_{1110} = 11010011.111_2$

$11010011.111_2 = \underbrace{011}_{3}\underbrace{010}_{2}\underbrace{011}_{3}.\underbrace{111}_{7} = 323.7_8$

$323.7_8 = 3*8^2 + 2*8^1 + 3*8^0 + 7*8^{-1} = 211.875_{10}$

（4） $1C3.F9_{16} = \underbrace{1}_{0001}\ \underbrace{C}_{1100}\ \underbrace{3}_{0011}\ .\ \underbrace{F}_{1111}\ \underbrace{9}_{1001} = 111000011.11111001_2$

$111000011.11111001_2 = \underbrace{111}_{7}\underbrace{000}_{0}\underbrace{011}_{3}.\underbrace{111}_{7}\underbrace{110}_{6}\underbrace{010}_{2} = 703.762_8$

$703.762_8 = 7*8^2 + 0*8^1 + 3*8^0 + 7*8^{-1} + 6*8^{-2} + 2*8^{-3} = 451.9726_{10}$

1-5

（1） $A(B+C) = AB + AC$

A	B	C	左式	右式
0	0	0	0	0
0	0	1	0	0
0	1	0	0	0
0	1	1	0	0
1	0	0	0	0
1	0	1	1	1
1	1	0	1	1
1	1	1	1	1

左式＝右式，得证。

（2） $A + BC = (A+B)(A+C)$

A	B	C	左式	右式
0	0	0	0	0
0	0	1	0	0
0	1	0	0	0
0	1	1	1	1
1	0	0	1	1
1	0	1	1	1
1	1	0	1	1
1	1	1	1	1

左式＝右式，得证。

（3）$\overline{A+B}=\overline{A}\,\overline{B}$

A	B	左式	右式
0	0	1	1
0	1	0	0
1	0	0	0
1	1	0	0

左式＝右式，得证。

（4）$\overline{AB}=\overline{A}+\overline{B}$

A	B	左式	右式
0	0	1	1
0	1	1	1
1	0	1	1
1	1	0	0

左式＝右式，得证。

（5）$A+\overline{BC}+\overline{A}BC=1$

A	B	C	左式	右式
0	0	0	1	1
0	0	1	1	1
0	1	0	1	1
0	1	1	1	1
1	0	0	1	1
1	0	1	1	1
1	1	0	1	1
1	1	1	1	1

左式＝右式，得证。

（6）$A\overline{B}+\overline{A}B=\overline{AB+\overline{A}\,\overline{B}}$

A	B	左式	右式
0	0	0	0
0	1	1	1
1	0	1	1
1	1	0	0

左式＝右式，得证。

（7）$A\oplus B=\overline{A}\oplus\overline{B}$

A	B	左式	右式
0	0	0	0
0	1	1	1
1	0	1	1
1	1	0	0

左式＝右式，得证。

（8）$A\overline{B}+B\overline{C}+C\overline{A}=\overline{A}B+\overline{B}C+\overline{C}A$

A	B	C	左式	右式
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	1	1
1	0	0	1	1
1	0	1	1	1
1	1	0	1	1
1	1	1	0	0

左式＝右式，得证。

1-6

（1）$A+\overline{A}B+\overline{B}=1$

证：$A+\overline{A}B+\overline{B}=A+B+\overline{B}=A+1=1$

（2）$A+B\overline{\overline{A}+CD}=A$

证：$A+B\overline{\overline{A}+CD}=A+\overline{\overline{A}}\,\overline{CD}=A+A\overline{CD}=A(1+\overline{CD})=A$

（3）$AB+\overline{A}C+\overline{B}C=AB+C$

证：$AB+\overline{A}C+\overline{B}C=AB+(\overline{A}+\overline{B})C=AB+\overline{AB}C=AB+C$

（4）$A\overline{B}+\overline{A+\overline{\overline{C}}}+\overline{B}(D+E)C=A\overline{B}+\overline{A}C$

证：$A\overline{B}+\overline{A+\overline{\overline{C}}}+\overline{B}(D+E)C=A\overline{B}+\overline{A}C+\overline{B}C(D+E)=A\overline{B}+\overline{A}C$

（5）$A?B\quad AB=A+B$

证：$A?B\quad AB=A\overline{B}+\overline{A}B+AB=A+\overline{A}B=A+B$

（6）$\overline{A\overline{B}+B\overline{C}+C\overline{A}}=\overline{A}\,\overline{B}\,\overline{C}+ABC$

证：$\overline{A\overline{B}+B\overline{C}+C\overline{A}}=(\overline{A}+B)(\overline{B}+C)(\overline{C}+A)=\overline{A}\,\overline{B}\,\overline{C}+ABC$

（7）$A\overline{B}\,\overline{D}+\overline{B}C\overline{D}+\overline{A}D+A\overline{B}C+\overline{A}\,\overline{B}C\overline{D}=A\overline{B}+\overline{A}D+\overline{B}C$

证：原式$=A\overline{B}\,\overline{D}+\overline{A}\,\overline{B}C\overline{D}+\overline{B}C\overline{D}+\overline{A}D+A\overline{B}C+\overline{A}\,\overline{B}C\overline{D}$ (再加一次最后一项)

$=\overline{B}\,\overline{D}(A+\overline{A}C)+\overline{B}C\overline{D}+\overline{A}D+\overline{B}C(A+\overline{A}\,\overline{D})$

$=\overline{B}\,\overline{D}(A+C)+\overline{B}C\overline{D}+\overline{A}D+\overline{B}C\overline{\overline{A}\,\overline{D}}$

$$= \overline{B}\overline{D}(A + C) + \overline{B}(C + \overline{C}D) + \overline{A}D$$

$$= A\overline{B}\overline{D} + \overline{B}(C\overline{D} + C + D) + \overline{A}D$$

$$= \overline{B}(A\overline{D} + D) + C\overline{B} + \overline{A}D$$

$$= (A\overline{B} + \overline{B}D + \overline{A}D) + C\overline{B}$$

$$= A\overline{B} + \overline{A}D + \overline{B}C$$

（8）$A \oplus B + B \oplus C + C \oplus D = A\overline{B} + B\overline{C} + C\overline{D} + D\overline{A}$

证：原式$= A\overline{B} + \overline{A}B + B\overline{C} + \overline{B}C + C\overline{D} + \overline{C}D$

$$= A\overline{B} + \overline{B}C + \overline{C}D + \overline{A}B + B\overline{C} + C\overline{D} + A\overline{D} + \overline{D}A$$

$$= A\overline{B} + B\overline{C} + C\overline{D} + D\overline{A} + \overline{A}B + \overline{B}C + \overline{C}D + A\overline{D}$$

$$= A\overline{B} + B\overline{C} + C\overline{D} + D\overline{A} + A\overline{B} + B\overline{C} + C\overline{D} + D\overline{A}$$

$$= A\overline{B} + B\overline{C} + C\overline{D} + D\overline{A}$$

1-7

（1）$F_1 = A\overline{B}C + \overline{A}B\overline{C}$

$$\overline{F}_1 = \left(\overline{A} + B + \overline{C}\right)(A + \overline{B} + C)$$

（2）$F_2 = A(\overline{B} + C) + \overline{C}(B + D)$

$$\overline{F}_2 = \left(\overline{A} + B\overline{C}\right)(C + \overline{B}\overline{D})$$

（3）$F_3 = (\overline{A} + B)(C + \overline{D})$

$$\overline{F}_3 = A\overline{B} + \overline{C}D$$

（4）$F_4 = (\overline{A}B + \overline{C}D)(B + A\overline{D})$

$$\overline{F}_4 = \left(A + \overline{B}\right)\left(\overline{C} + D\right) + \overline{B}\left(\overline{A} + D\right)$$

（5）$F_5 = A\overline{B} + \overline{A}C\overline{B + D}$

$$\overline{F}_5 = \left(\overline{A} + B\right)\left(A + \overline{C} + \overline{\overline{B}\overline{D}}\right)$$

（6）$F_6 = \overline{A + B\overline{C}} + \overline{\overline{B} + \overline{C}D}$

$\overline{F_6} = \overline{\overline{A}\left(\overline{B}+C\right)}\overline{B\overline{C}+\overline{D}}$

（7） $F_7 = \overline{A\overline{C}+BD}\overline{C}+\overline{A+\overline{BD}}$

$\overline{F_7} = \overline{\left(\overline{A}+C\right)\left(\overline{B}+\overline{D}\right)}+\overline{C\overline{A}\overline{B}+\overline{D}}$

（8） $F_8 = \overline{(A+\overline{D})(\overline{B}+C)}+\overline{(\overline{A+C}+B)\overline{AB+CD}}$

$\overline{F_8} = \overline{\overline{A}D+B\overline{C}}\,\overline{\overline{A}\overline{C}\overline{B}+\left(\overline{A}+\overline{B}\right)\left(\overline{C}+\overline{D}\right)}$

1-8

（1） $F_1 = A\overline{B}+\overline{C}D$

$F_1^{'} = (A+\overline{B})(\overline{C}+D)$

（2） $F_2 = (A+\overline{B})(\overline{C}+D)$

$F_2^{'} = A\overline{B}+\overline{C}D$

（3） $F_3 = \overline{A}(B+\overline{D})+B(A+\overline{C})$

$F_3^{'} = (\overline{A}+B\overline{D})(B+A\overline{C})$

（4） $F_4 = (A+B\overline{C}D)(\overline{A}BC+\overline{D})$

$F_4^{'} = A(B+\overline{C}+D)+(\overline{A}+B+C)\overline{D}$

（5） $F_5 = \overline{A+B}+\overline{C+D}$

$F_5^{'} = \overline{AB}\,\overline{CD}$

（6） $F_6 = \overline{B\overline{C}+C\overline{D}}\,\overline{B(\overline{A}D+\overline{C})}$

$F_6^{'} = \overline{(B+\overline{C})(C+\overline{D})}+\overline{\overline{B}+(\overline{A}+D)\overline{C}}$

（7） $F_7 = \overline{BC+AD}\,\overline{A\overline{C}+\overline{C+\overline{AB}}}$

$F_7^{'} = \overline{(B+C)(A+D)}+\overline{\overline{A+C}\,\overline{C}\,\overline{A+B}}$

（8） $F_8 = ABC+\overline{A+C\overline{D}(B\overline{D}+C)}+(\overline{B}C+\overline{A+D})\overline{B+\overline{A+\overline{B}C}}$

$F_8^{'} = (A+B+C)\overline{A(C+\overline{D})+(B+\overline{D})C}(\overline{B}+C)\overline{AD}+\overline{B\overline{A(\overline{B}+C)}}$

1-9

（1）$F_1 = A\overline{B}C + \overline{A}BC + \overline{A}\overline{B}\overline{C} + ABC$

A	B	C	F
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

A \ BC	00	01	11	10
0	1	0	1	0
1	0	1	1	0

（2）$F_2 = A + BC + CD$

A	B	C	D	F	A	B	C	D	F
0	0	0	0	0	1	0	0	0	1
0	0	0	1	0	1	0	0	1	1
0	0	1	0	0	1	0	1	0	1
0	0	1	1	1	1	0	1	1	1
0	1	0	0	0	1	1	0	0	1
0	1	0	1	0	1	1	0	1	1
0	1	1	0	1	1	1	1	0	1
0	1	1	1	1	1	1	1	1	1

AB \ CD	00	01	11	10
00	0	0	1	0
01	0	0	1	1
11	1	1	1	1
10	1	1	1	1

（3）$F_3 = \overline{A}B + \overline{B}(C + AD)$

A	B	C	D	F
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1
0	1	0	0	1
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1

A	B	C	D	F
1	0	0	0	0
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

AB \ CD	00	01	11	10
00	0	0	1	1
01	1	1	1	1
11	0	0	0	0
10	0	1	1	1

（4） $F_4 = (A + \overline{B} + C)(\overline{A} + B + C)(\overline{A} + B + \overline{C})$

A	B	C	F
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

A \ BC	00	01	11	10
0	1	1	1	0
1	0	0	1	1

（5） $F_5 = (\overline{B}D + C)(C + AD)$

A	B	C	D	F	A	B	C	D	F
0	0	0	0	0	1	0	0	0	0
0	0	0	1	0	1	0	0	1	1
0	0	1	0	1	1	0	1	0	1
0	0	1	1	1	1	0	1	1	1
0	1	0	0	0	1	1	0	0	0
0	1	0	1	0	1	1	0	1	0
0	1	1	0	1	1	1	1	0	1
0	1	1	1	1	1	1	1	1	1

AB \ CD	00	01	11	10
00	0	0	1	1
01	0	0	1	1
11	0	0	1	1
10	0	1	1	1

（6） $F_6 = (A\overline{B} + C\overline{D})(B\overline{C} + D\overline{A})(\overline{A}C + B\overline{D})$

A	B	C	D	F	A	B	C	D	F
0	0	0	0	0	1	0	0	0	0
0	0	0	1	0	1	0	0	1	0
0	0	1	0	0	1	0	1	0	0
0	0	1	1	0	1	0	1	1	0
0	1	0	0	0	1	1	0	0	0
0	1	0	1	0	1	1	0	1	0
0	1	1	0	0	1	1	1	0	0
0	1	1	1	0	1	1	1	1	0

AB \ CD	00	01	11	10
00	0	0	0	0
01	0	0	0	0
11	0	0	0	0
10	0	0	0	0

（7） $F_7 = A + \overline{B}\overline{C} + B\overline{A + \overline{C} + \overline{D}}$

A	B	C	D	F	A	B	C	D	F
0	0	0	0	1	1	0	0	0	1
0	0	0	1	1	1	0	0	1	1
0	0	1	0	0	1	0	1	0	1
0	0	1	1	0	1	0	1	1	1
0	1	0	0	0	1	1	0	0	1
0	1	0	1	0	1	1	0	1	1
0	1	1	0	0	1	1	1	0	1
0	1	1	1	1	1	1	1	1	1

AB \ CD	00	01	11	10
00	1	1	0	0
01	0	0	1	0
11	1	1	1	1
10	1	1	1	1

（8）$F_8 = \overline{A}B\overline{D} + \overline{B}C + \overline{C}(A+D) + \overline{D}\,\overline{A(C+\overline{BD})}$

A	B	C	D	F	A	B	C	D	F
0	0	0	0	1	1	0	0	0	1
0	0	0	1	1	1	0	0	1	1
0	0	1	0	1	1	0	1	0	1
0	0	1	1	1	1	0	1	1	1
0	1	0	0	1	1	1	0	0	1
0	1	0	1	1	1	1	0	1	1
0	1	1	0	1	1	1	1	0	0
0	1	1	1	0	1	1	1	1	0

AB \ CD	00	01	11	10
00	1	1	1	1
01	1	1	0	1
11	1	1	0	0
10	1	1	1	1

1-10

标准与或式：

$$F = \overline{A}\,\overline{B}\,\overline{C}D + \overline{A}\,\overline{B}C\overline{D} + \overline{A}B\overline{C}\,\overline{D} + \overline{A}BCD + A\overline{B}\,\overline{C}\,\overline{D} + A\overline{B}C\overline{D} + ABC\overline{D}$$

标准或与式：

$$F=(A+B+C+D)(A+B+\overline{C}+D)(A+\overline{B}+C+\overline{D})(A+\overline{B}+\overline{C}+D)$$
$$(\overline{A}+B+C+\overline{D})(\overline{A}+B+\overline{C}+\overline{D})(\overline{A}+\overline{B}+C+D)(\overline{A}+\overline{B}+C+\overline{D})$$
$$(\overline{A}+\overline{B}+\overline{C}+D)$$

1-11

(1) $F_1=\sum(0,2,4,5,6,7)$

(2) $F_2=\sum(0,2,3,4,5,12,13,15)$

(3) $F_3=\sum(0,1,2,4,5)$

(4) $F_4=\sum(2,4,5,6,7,14,15)$

(5) $F_5=\sum(0,3,4,5,6)$

(6) $F_6=\sum(0,1,3,4,5,8,10,11,12,13,14,15)$

(7) $F_7=\sum(0,2,5,6)$

(8) $F_8=\sum(1,4,5,7,11,14)$

1-12

(1) $F_1=\prod(0,7)$

(2) $F_2=\prod(2,3,5,6,7,13,15)$

(3) $F_3=\prod(6,7,11,14,15)$

(4) $F_4=\prod(0,1,2,3,4,6,7,10,12,14)$

(5) $F_5=\prod(3,4,5,7)$

(6) $F_6=\prod(0,1,5,8,9,10,11,13)$

(7) $F_7=\prod(2,3,5,7)$

(8) $F_8=\prod(0,2,4,7,9,12,15)$

1-13

(a) $F=\sum(1,3,6,10,11,12)$

$F=\prod(0,2,4,5,7,8,9,13,14,15)$

(b) $F=\sum(0,3,4,6,8,9,13,15)$

$$F=\prod(1,2,5,7,10,11,12,14)$$

1-14

(1) $A+\overline{A}B+BC\overline{D}=A+B+BC\overline{D}=A+B$

(2) $AB+\overline{B}\overline{C}+\overline{A}+C=\overline{A}+B+C+\overline{B}=1$

(3) $\overline{A}+B+\overline{A\overline{B}}(C+D)=\overline{A\overline{B}}+\overline{A\overline{B}}(C+D)=\overline{A}+B$

(4)
$$\begin{aligned}&\overline{A}\overline{B}+AC+B\overline{C}+A?B\\&=\overline{A}\overline{B}+AC+B\overline{C}+\overline{A}B+A\overline{B}\\&=\overline{A}\overline{B}+\overline{A}B+(AC+A\overline{B})+B\overline{C}\\&=\overline{A}+A+B\overline{C}=1\end{aligned}$$

(5)
$$\begin{aligned}&BD+ABC\overline{D}+\overline{\overline{A}+B+\overline{C}}\\&=BD+ABC\overline{D}+A\overline{B}C\\&=BD+AC(\overline{D}+\overline{B})\\&=BD+AC\end{aligned}$$

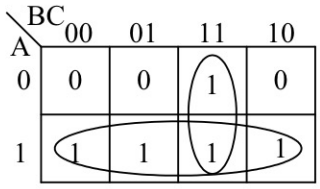
(6)
$$\begin{aligned}&A\overline{B}+(A+C)(D+\overline{A+B})+(\overline{C}+\overline{D})E\\&=A\overline{B}+AD+CD+\overline{A}\overline{B}C+\overline{CD}E\\&=A\overline{B}+AD+CD+\overline{B}C+E\end{aligned}$$

(7)
$$\begin{aligned}&C+A\overline{B}C\overline{D}+\overline{C}\overline{D}+(\overline{A}+B)(CD+\overline{C}\overline{D})\\&=C+A\overline{B}(CD+\overline{C}\overline{D})+\overline{A\overline{B}}(CD+\overline{C}\overline{D})\\&=C+CD+\overline{C}\overline{D}=C+\overline{D}\end{aligned}$$

(8) $A\overline{B}+B\overline{C}+C\overline{A}+\overline{\overline{A}B}=A\overline{B}+B\overline{C}+C\overline{A}+A+\overline{B}=A+\overline{C}+\overline{B}+C=1$

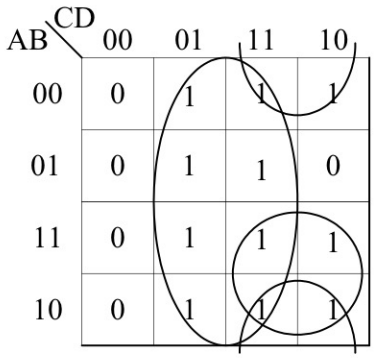
1-15

（1）$A\overline{B}+BC+A\overline{C}$



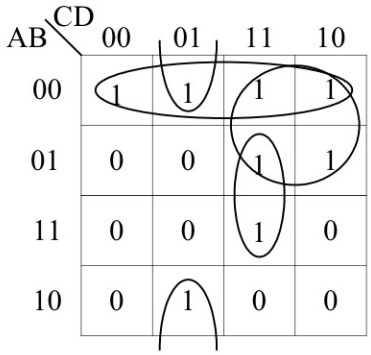
$A\overline{B}+BC+A\overline{C}=A+BC$

（2）$ABC+\overline{B}C+\overline{A}D+\overline{C}D$



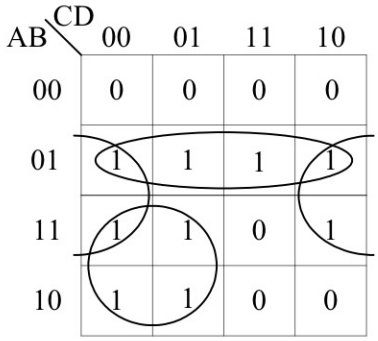
$ABC+\overline{B}C+\overline{A}D+\overline{C}D=\overline{B}C+AC+D$

（3）$\overline{A}\overline{B}\overline{C}+A\overline{B}\overline{C}D+ABCD+\overline{A}C$



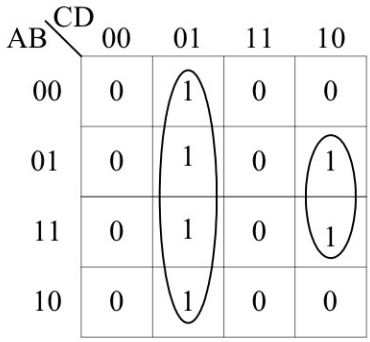
$\overline{A}\overline{B}\overline{C}+A\overline{B}\overline{C}D+ABCD+\overline{A}C=\overline{A}\overline{B}+\overline{A}C+\overline{B}\overline{C}D+BCD$

（4）$B(\overline{C}+\overline{AD})+A\overline{C(\overline{B}+D)}$



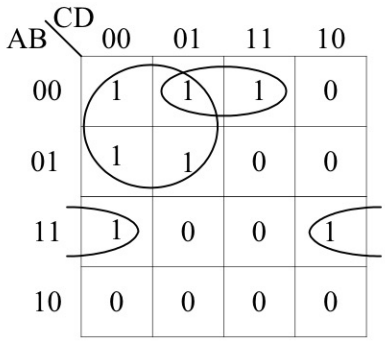
$B(\overline{C}+\overline{AD})+\overline{A\overline{C(\overline{B}+D)}}=\overline{A}B+A\overline{C}+B\overline{D}$

（5）$B(C?\ D)\quad A\overline{C}D+\overline{B}\overline{C}D+BC\overline{D}$



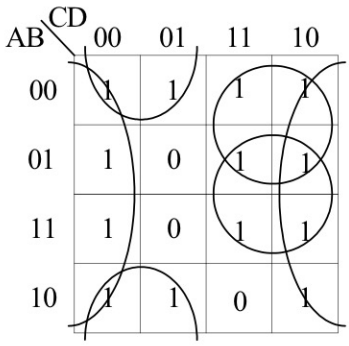
$B(C?\ D)\quad A\overline{C}D+\overline{B}\overline{C}D+BC\overline{D}=\overline{C}D+BC\overline{D}$

（6）$\overline{\overline{A}BC+\overline{B}C\overline{D}+AD+A(\overline{B}+\overline{C}D)}$



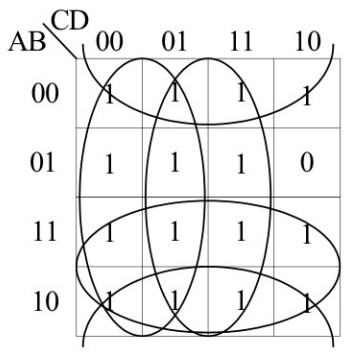
$\overline{\overline{A}BC+\overline{B}C\overline{D}+AD+A(\overline{B}+\overline{C}D)}=AB\overline{D}+\overline{A}\overline{B}D+\overline{A}\overline{C}$

（7）$\overline{A\overline{B}C+\overline{CD+AC}+\overline{BD+ACD}}$



$\overline{A\overline{B}C+\overline{CD+AC}}+\overline{BD+ACD}=\overline{A}C+BC+\overline{B}\overline{C}+\overline{D}$

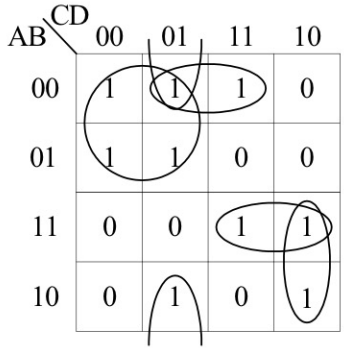
（8）$\overline{ACD+\overline{A}BC\overline{D}+\overline{A}\overline{B}(C+\overline{D})+D}$



$$\overline{ACD+\overline{A}BC\overline{D}+\overline{A}\overline{B}(C+\overline{D})+D}=A+\overline{B}+\overline{C}+D$$

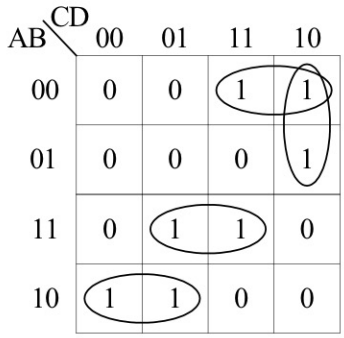
1-16

（1）$F_1(A,B,C,D)=\sum m(0,1,3,4,5,9,10,14,15)$



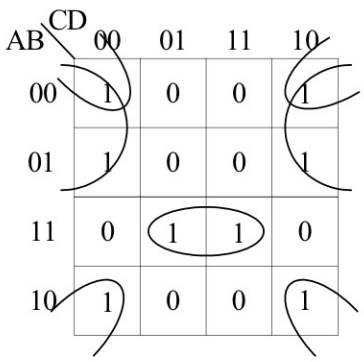
$$F_1(A,B,C,D)=\overline{A}\overline{C}+\overline{A}\overline{B}D+\overline{B}\overline{C}D+ABC+AC\overline{D}$$

（2）$F_2(A,B,C,D)=\sum m(2,3,6,8,913,15)$



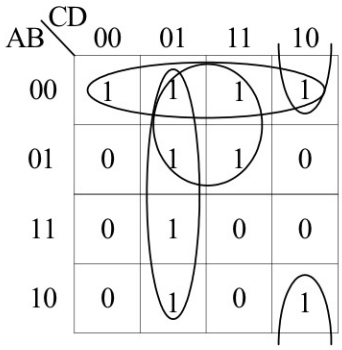
$$F_2(A,B,C,D)=\overline{A}\overline{B}C+\overline{A}C\overline{D}+ABD+A\overline{B}\overline{C}$$

（3）$F_3(A,B,C,D)=\sum m(0,2,4,6,8,10,13,15)$



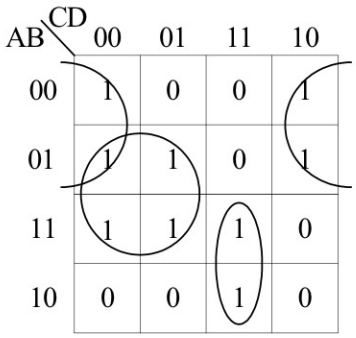
$F_3(A,B,C,D)=\overline{B}\overline{D}+\overline{A}\overline{D}+ABD$

（4）$F_4(A,B,C,D)=\sum m(0,1,2,3,5,7,9,10,13)$



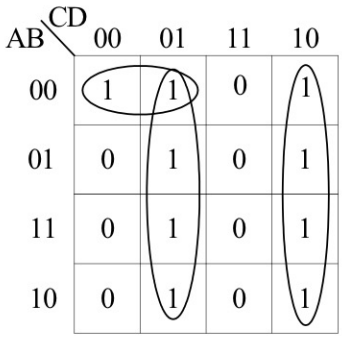
$F_4(A,B,C,D)=\overline{A}\overline{B}+\overline{C}D+\overline{B}\overline{D}C+\overline{A}D$

（5）$F_5(A,B,C,D)=\prod M(1,3,7,8,9,10,14)$



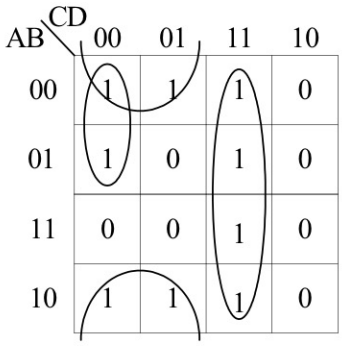
$F_5(A,B,C,D)=\overline{A}\overline{D}+B\overline{C}+ACD$

（6）$F_6(A,B,C,D)=\prod M(3,4,7,8,11,12,15)$



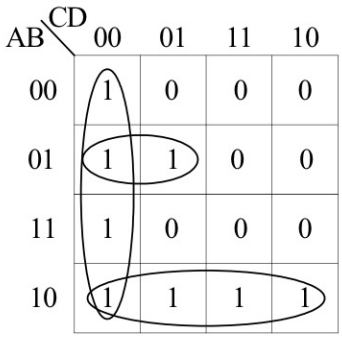
$F_6(A,B,C,D)=\overline{C}D+C\overline{D}+\overline{A}\overline{B}\overline{C}$

（7） $F_7(A,B,C,D)=\sum M(2,5,6,10,12,13,14)$



$F_7(A,B,C,D)=CD+\overline{B}\overline{C}+\overline{A}\overline{C}\overline{D}$

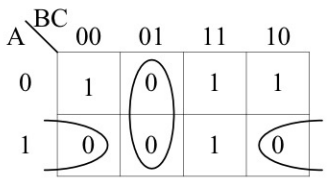
（8） $F_8(A,B,C,D)=\sum M(1,2,3,6,7,13,14,15)$



$F_8(A,B,C,D)=\overline{C}\overline{D}+A\overline{B}+\overline{A}B\overline{C}$

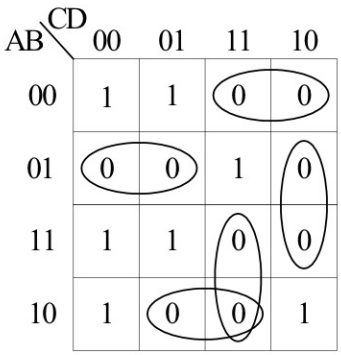
1-17

（1） $F_1(A,B,C)=\sum m(0,2,3,7)$



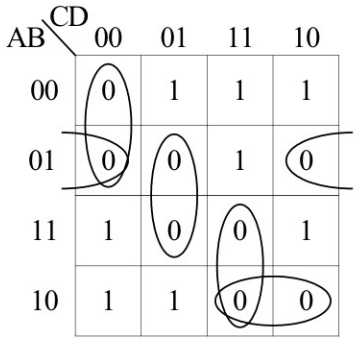
$$F_1(A,B,C) = (B + \overline{C})(\overline{A} + C)$$

（2） $F_2(A,B,C,D) = \sum m(0,1,7,8,10,12,13)$



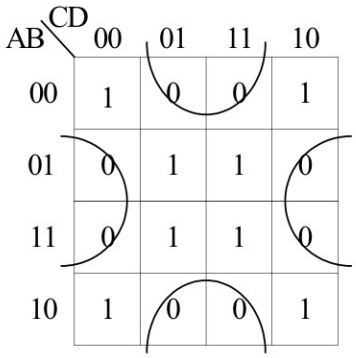
$$F_2(A,B,C,D) = (A + \overline{B} + C)(A + B + \overline{C})(\overline{B} + \overline{C} + D)(\overline{A} + \overline{C} + \overline{D})(\overline{A} + B + \overline{D})$$

（3） $F_3(A,B,C,D) = \sum m(1,2,3,7,8,9,12,14)$



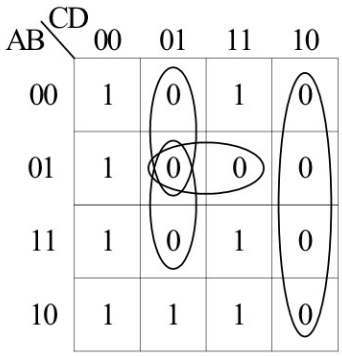
$$F_3(A,B,C,D) = (A + C + D)(A + \overline{B} + D)(\overline{B} + C + \overline{D})(\overline{A} + \overline{C} + \overline{D})(\overline{A} + B + \overline{C})$$

（4） $F_4(A,B,C,D) = \sum m(0,2,5,7,8,10,13,15)$



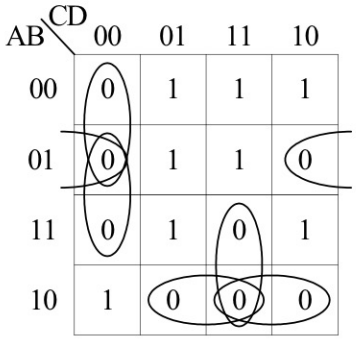
$F_4(A,B,C,D) = (B + \overline{D})(\overline{B} + D)$

（5） $F_5(A,B,C,D) = \tilde{O} M(1,2,5,6,7,10,13,14)$



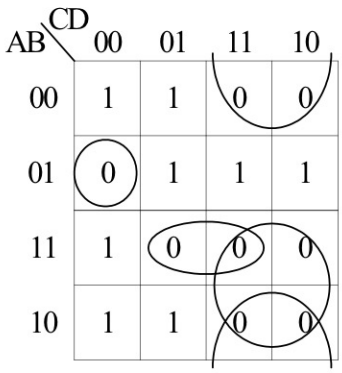
$F_5(A,B,C,D) = (\overline{C} + D)(A + C + \overline{D})(\overline{B} + C + \overline{D})(A + \overline{B} + \overline{D})$

（6） $F_6(A,B,C,D) = \tilde{O} M(0,4,6,9,10,11,12,15)$



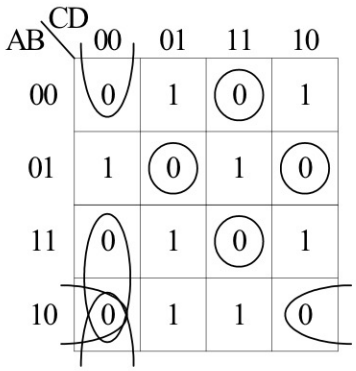
$F_6(A,B,C,D) = (A + C + D)(\overline{B} + C + D)(A + \overline{B} + D)(\overline{A} + B + \overline{D})(\overline{A} + B + \overline{C})(\overline{A} + \overline{C} + \overline{D})$

（7） $F_7(A,B,C,D) = \tilde{O} M(2,3,4,10,11,13,14,15)$



$$F_7(A,B,C,D)=(A+\overline{B}+C+D)(B+\overline{C})(\overline{A}+\overline{C})(\overline{A}+\overline{B}+\overline{D})$$

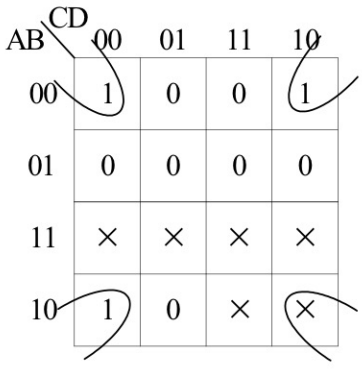
（8） $F_8(A,B,C,D)=\tilde{\Omega}M(0,3,5,6,8,10,12,15)$



$$F_8(A,B,C,D)=(B+C+D)(\overline{A}+C+D)(\overline{A}+B+D)(A+B+\overline{C}+\overline{D})$$
$$=(A+\overline{B}+C+\overline{D})(A+\overline{B}+\overline{C}+D)(\overline{A}+\overline{B}+\overline{C}+\overline{D})$$

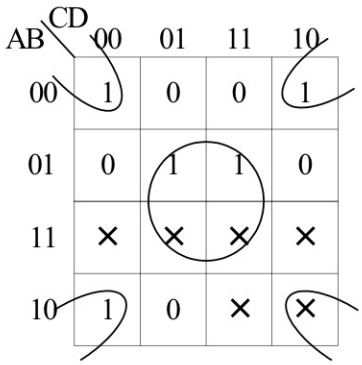
1-18

（1） $F_1=\overline{A}\overline{B}\overline{C}\overline{D}+\overline{A}\overline{B}C\overline{D}+A\overline{B}\overline{C}\overline{D}$　　约束条件 $AB+AC=0$



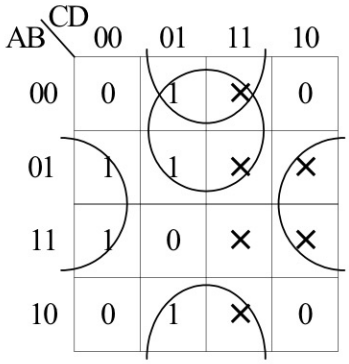
$$F_1=\overline{B}\overline{D}$$

（2）$F_2=\overline{A}BD+\overline{A}\overline{B}\overline{D}+\overline{B}\overline{C}\overline{D}$　　约束条件 $AB+AC=0$



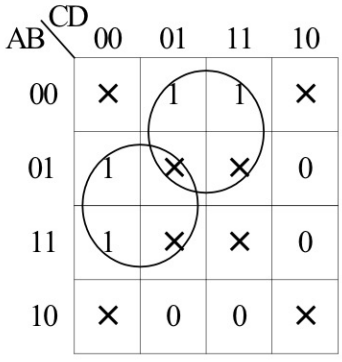
$F_2=\overline{B}\overline{D}+BD$

（3）$F_3=B\overline{C}\overline{D}+\overline{B}\overline{C}D+\overline{A}B\overline{C}D$　　约束条件 $BC+CD=0$



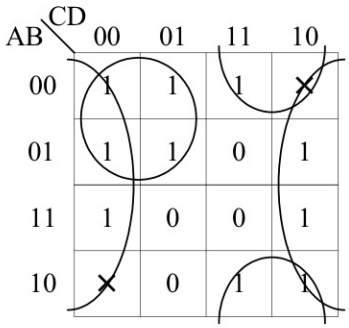
$F_3=B\overline{D}+\overline{A}D+\overline{B}D$

（4）$F_4=\overline{A}\overline{B}CD+B\overline{C}\overline{D}+\overline{A}\overline{B}CD$　　约束条件 $BD+\overline{B}\overline{D}=0$



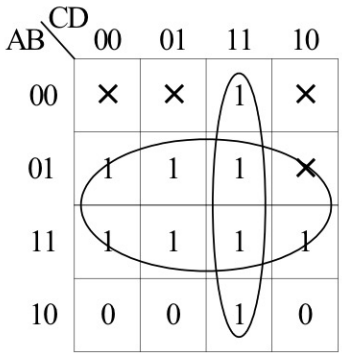
$F_4=B\overline{C}+\overline{A}D$

（5）$F_5=\overline{A}\overline{C}+B\overline{D}+A\overline{B}C+\overline{A}BCD$　　约束条件 $A\overline{B}\overline{C}\overline{D}+\overline{A}\overline{B}C\overline{D}=0$



$F_5 = \overline{D} + \overline{B}C + \overline{AC}$

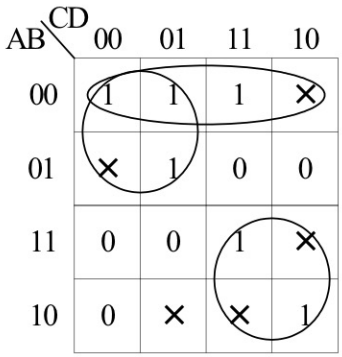
（6）$F_6 = AB + B\overline{C} + CD$　　约束条件 $d(0,1,2,6) = 0$



$F_6 = B + CD$

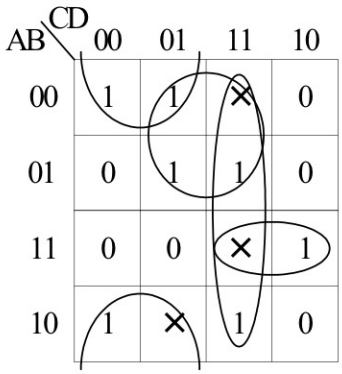
1-19

（1）$F_1(A,B,C,D) = \sum m(0,1,3,5,10,15) + d(2,4,9,11,14)$



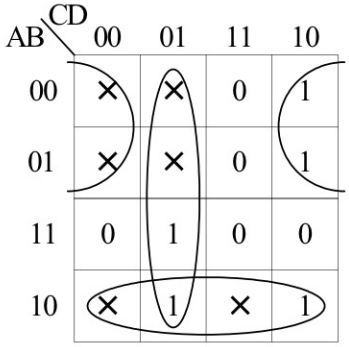
$F_1(A,B,C,D) = \overline{AB} + AC + \overline{AC}$

（2）$F_2(A,B,C,D) = \sum m(0,1,5,7,8,11,14) + d(3,9,15)$



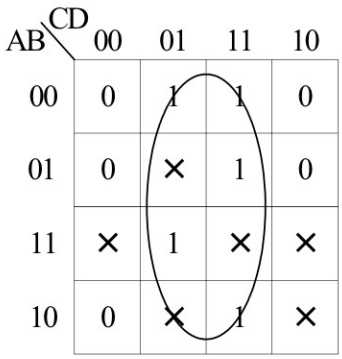
$F_2(A,B,C,D) = \overline{B}\overline{C} + \overline{A}D + CD + ABC$

（3） $F_3(A,B,C,D) = \sum m(2,6,9,10,13) + d(0,1,4,5,8,11)$



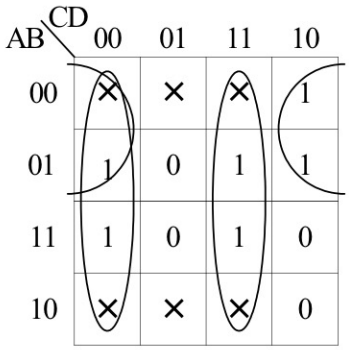
$F_3(A,B,C,D) = \overline{A}\overline{D} + \overline{C}D + A\overline{B}$

（4） $F_4(A,B,C,D) = \sum m(1,3,7,11,13) + d(5,9,10,12,14,15)$



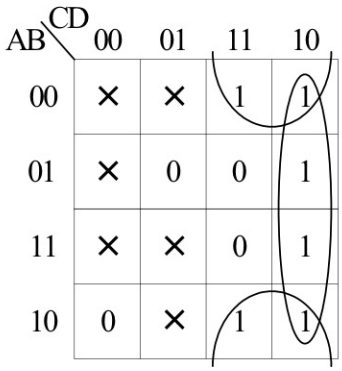
$F_4(A,B,C,D) = D$

（5） $F_5(A,B,C,D) = \sum m(2,4,6,7,12,15) + d(0,1,3,8,9,11)$



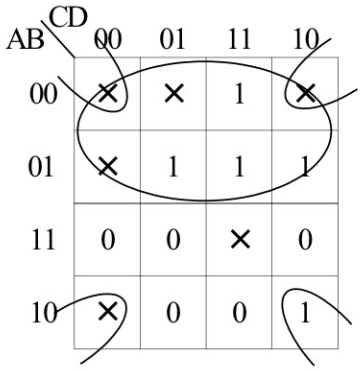
$F_5(A,B,C,D)=\overline{C}\overline{D}+CD+\overline{A}\overline{D}$

（6）$F_6(A,B,C,D)=\sum m(2,3,6,10,11,14)+\quad d(0,1,4,9,12,13)$



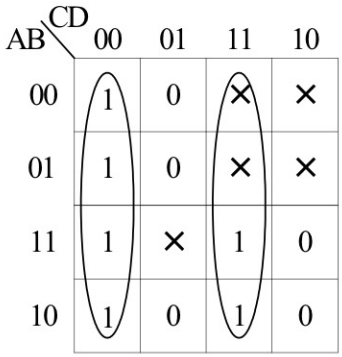
$F_6(A,B,C,D)=C\overline{D}+\overline{B}C$

（7）$F_7(A,B,C,D)=\sum m(3,5,6,7,10)+\quad d(0,1,2,4,8,15)$



$F_7(A,B,C,D)=\overline{A}+\overline{B}\overline{D}$

（8）$F_8(A,B,C,D)=\sum m(0,4,8,11,12,15)+\quad d(2,3,6,7,13)$



$$F_8(A,B,C,D) = CD + \overline{C}\,\overline{D}$$

第二章习题

2-1

a) $Z_1 = AB + BC = B(A + C)$

真值表:

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

b) $Z_2 = \overline{A + BC + D} = \overline{A}\,\overline{B}\overline{C}\,\overline{D}$

真值表：

A	B	C	D	F	A	B	C	D	F
0	0	0	0	1	1	0	0	0	0
0	0	0	1	0	1	0	0	1	0
0	0	1	0	0	1	0	1	0	0
0	0	1	1	0	1	0	1	1	0
0	1	0	0	0	1	1	0	0	0
0	1	0	1	0	1	1	0	1	0
0	1	1	0	0	1	1	1	0	0
0	1	1	1	0	1	1	1	1	0

c) $Z_3 = \overline{\overline{C + D} + B + A} \,?\, E \quad \overline{\overline{C + D} + B + A} \odot E = \overline{(C + D)\overline{B} + A} \odot E$

真值表：

A	B	C	D	E	F
0	0	0	0	0	0
0	0	0	0	1	1
0	0	0	1	0	1
0	0	0	1	1	0
0	0	1	0	0	1
0	0	1	0	1	0
0	0	1	1	0	1
0	0	1	1	1	0

A	B	C	D	E	F
0	1	0	0	0	0
0	1	0	0	1	1
0	1	0	1	0	0
0	1	0	1	1	1
0	1	1	0	0	0
0	1	1	0	1	1
0	1	1	1	0	0
0	1	1	1	1	1

A	B	C	D	E	F
1	0	0	0	0	1
1	0	0	0	1	0
1	0	0	1	0	1
1	0	0	1	1	0
1	0	1	0	0	1
1	0	1	0	1	0
1	0	1	1	0	1
1	0	1	1	1	0

A	B	C	D	E	F
1	1	0	0	0	1
1	1	0	0	1	0
1	1	0	1	0	1
1	1	0	1	1	0
1	1	1	0	0	1
1	1	1	0	1	0
1	1	1	1	0	1
1	1	1	1	1	0

2-2

a)表达式：$Z = \overline{\overline{B + C}\overline{AC}\overline{A}}$？ $\overline{B \quad C}$？ $\overline{A} \quad \overline{A}\overline{B}\overline{C}$ 排$\left(\overline{B}\overline{C}\right) \quad \overline{A}$

真值表：

A	B	C	F
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	0

b)表达式：$X = B$ 排$C \quad A$

$Y = \overline{A}B + BC + \overline{A}C$

真值表：

A	B	C	X	Y
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

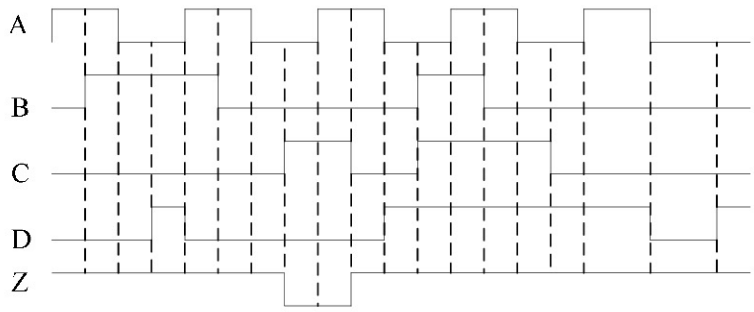
2-3

表达式：$Z = \overline{\overline{ADB}\,\overline{BCD}\,\overline{ADC}} = \overline{\overline{BC}\,\overline{D}}$

真值表：

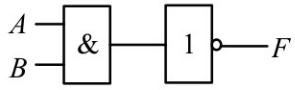
A	B	C	D	Z	A	B	C	D	Z
0	0	0	0	1	1	0	0	0	1
0	0	0	1	1	1	0	0	1	1
0	0	1	0	0	1	0	1	0	0
0	0	1	1	1	1	0	1	1	1
0	1	0	0	1	1	1	0	0	1
0	1	0	1	1	1	1	0	1	1
0	1	1	0	1	1	1	1	0	1
0	1	1	1	1	1	1	1	1	1

波形图：

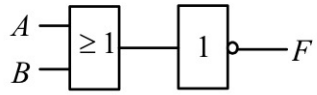


2-4

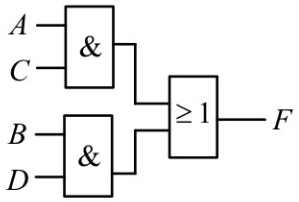
1)



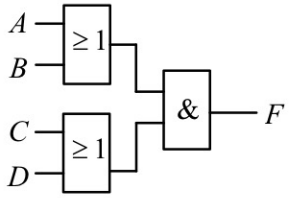
2)



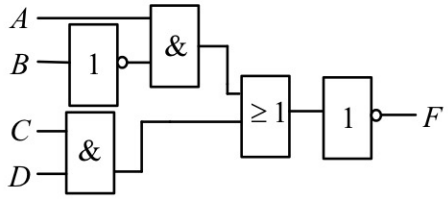
3)



4)

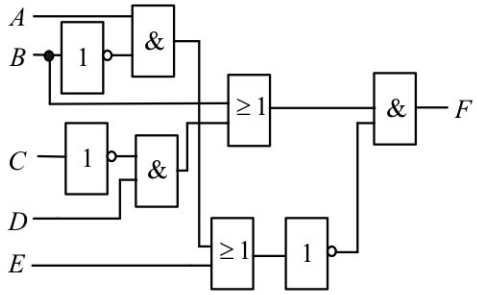


5)



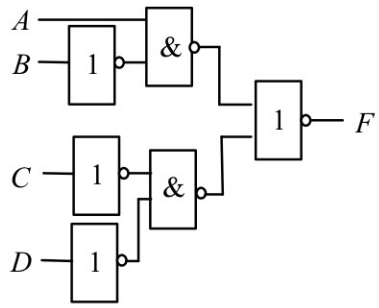
6)

$$F_6 = \left(B + \overline{C}D\right)\overline{A\overline{B} + \overline{E}} = \overline{E}B + \left(\overline{A} + B\right)\overline{C}DE$$

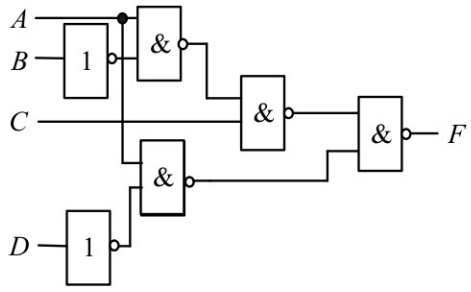


2-5

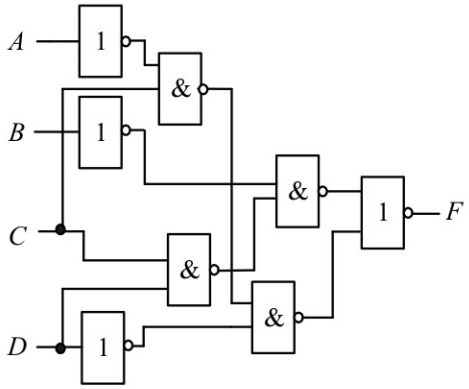
1) $F=\overline{\overline{A}\overline{B}\overline{C}\overline{D}}$



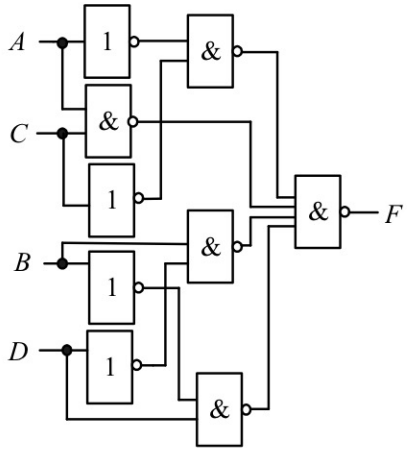
2) $F=\overline{\overline{A}\overline{B}\overline{C}}\,\overline{A\overline{D}}$



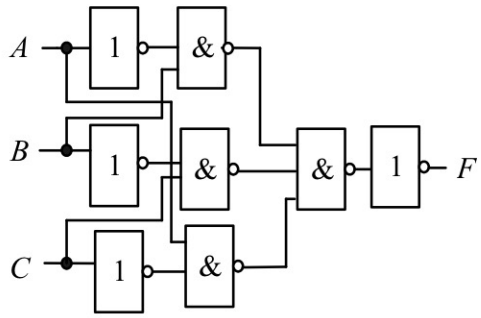
3) $F=\overline{\overline{A}\overline{C}\overline{D}}\,\overline{B\overline{C}\overline{D}}$



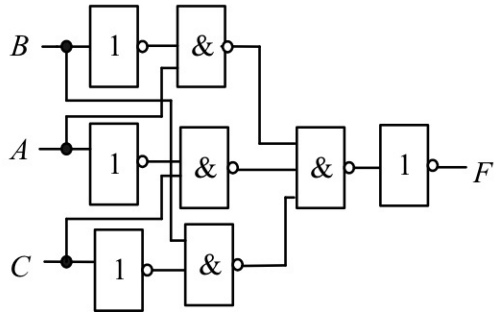
4) $F = \overline{\overline{AC}\,\overline{AC}\,\overline{BD}\,\overline{BD}}$



5) $F = \overline{\overline{AB}\,\overline{BC}\,\overline{CA}}$

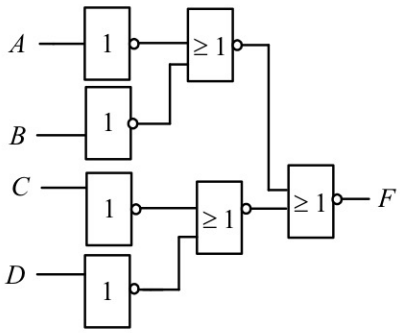


6) $F = \overline{\overline{AB}\,\overline{BC}\,\overline{CA}}$

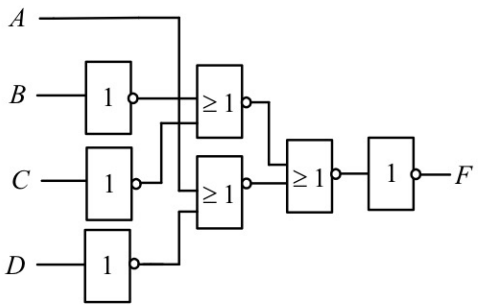


2-6

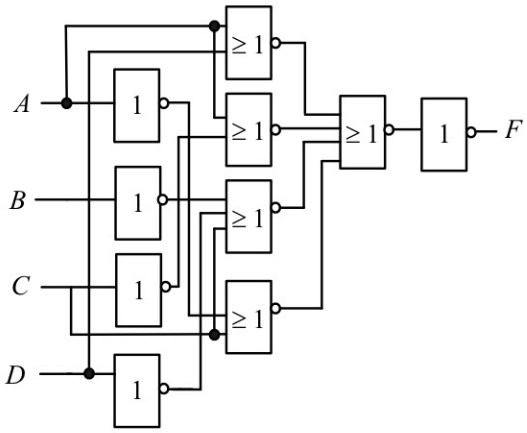
1) $F = \overline{\overline{A} + \overline{B} + \overline{C} + \overline{D}}$



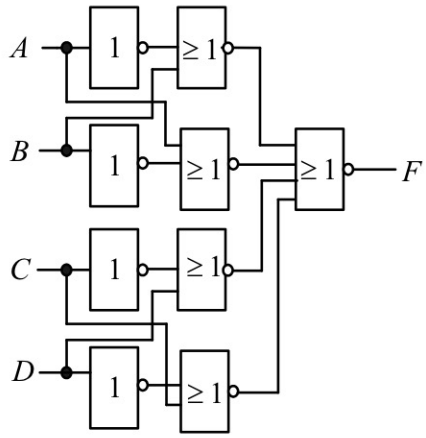
2) $F = \overline{\overline{A} + \overline{D} + \overline{B} + \overline{C}}$



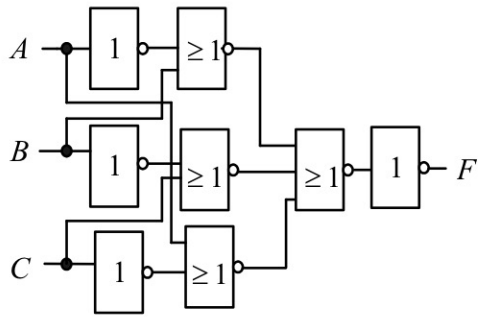
3) $F = \overline{\overline{\overline{A} + C} + \overline{\overline{B} + \overline{D}} + C} + \overline{\overline{A} + \overline{C}} + \overline{A + D}$



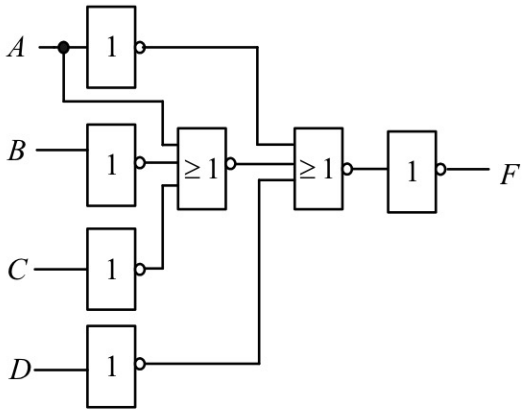
4) $F=\overline{\overline{\overline{A}+B}+\overline{A+\overline{B}}+\overline{\overline{C}+D}+\overline{C+\overline{D}}}$



5) $F=\overline{\overline{\overline{A}+B}+\overline{\overline{B}+C}+\overline{\overline{C}+A}}$



6) $F=\overline{A+\overline{B}+\overline{\overline{C}+\overline{A}+\overline{D}}}$



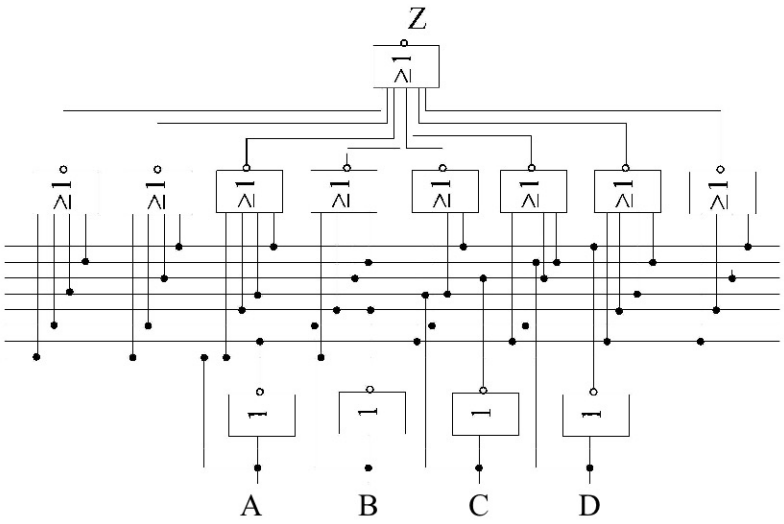
2-7

（1）卡诺图及表达式：

AB \ CD	00	01	11	10
00	0	1	0	1
01	1	0	1	0
11	0	1	0	1
10	1	0	1	0

$$F=(A+B+C+D)(A+B+\overline{C}+\overline{D})(A+\overline{B}+C+\overline{D})(A+\overline{B}+\overline{C}+D)$$
$$(\overline{A}+\overline{B}+C+D)(\overline{A}+\overline{B}+\overline{C}+\overline{D})(\overline{A}+B+C+\overline{D})(\overline{A}+B+\overline{C}+D)$$

（2）电路图：



2-8

（1）真值表：

S_2	S_1	S_0	A	B	F
0	0	0	0	0	1
0	0	0	0	1	0
0	0	0	1	0	0
0	0	0	1	1	0
0	0	1	0	0	0
0	0	1	0	1	1
0	0	1	1	0	0
0	0	1	1	1	0
0	1	0	0	0	0
0	1	0	0	1	0
0	1	0	1	0	1
0	1	0	1	1	0
0	1	1	0	0	0
0	1	1	0	1	0
0	1	1	1	0	0
0	1	1	1	1	1

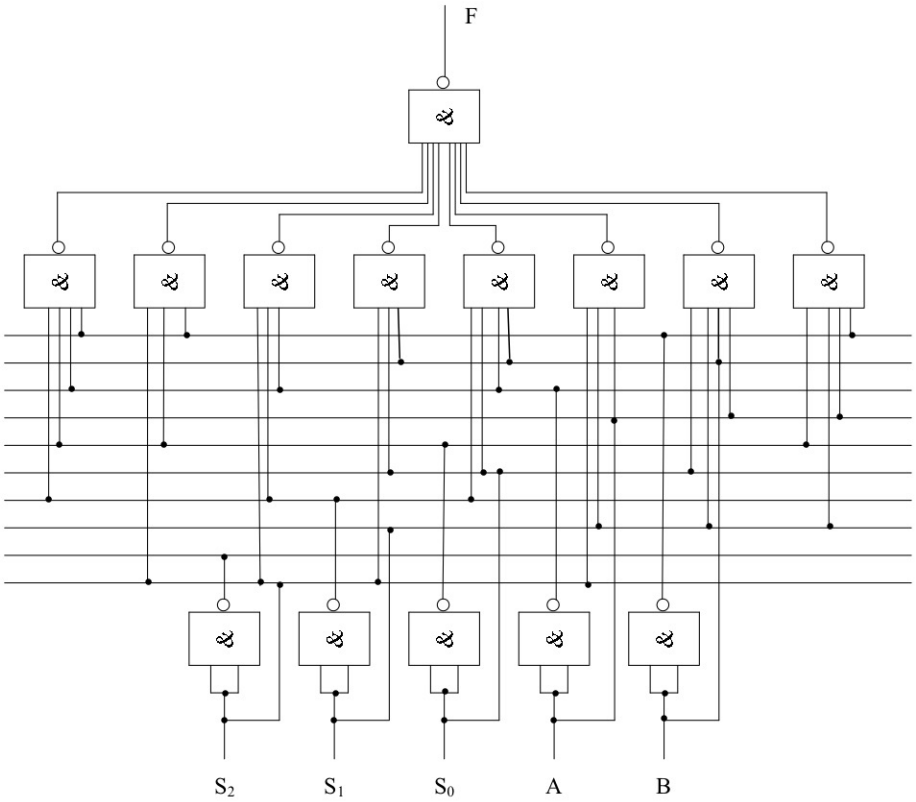
S_2	S_1	S_0	A	B	F
1	0	0	0	0	1
1	0	0	0	1	1
1	0	0	1	0	1
1	0	0	1	1	0
1	0	1	0	0	1
1	0	1	0	1	1
1	0	1	1	0	0
1	0	1	1	1	1
1	1	0	0	0	1
1	1	0	0	1	0
1	1	0	1	0	1
1	1	0	1	1	1
1	1	1	0	0	0
1	1	1	0	1	0
1	1	1	1	0	0
1	1	1	1	1	1

（2）卡诺图及表达式：

$S_2S_1S_0$ / AB	000	001	011	010	110	111	101	100
00	1	0	0	0	1	0	1	1
01	0	1	0	0	0	1	1	1
11	0	0	1	0	1	1	1	0
10	0	0	0	1	1	1	0	1

$$F=\overline{S_1}\overline{S_0}\overline{A}\overline{B}+S_2\overline{S_0}\overline{B}+\overline{A}S_2\overline{S_1}+BS_2S_0+\overline{A}B\overline{S_1}S_0+ABS_1S_0+AS_2S_1+A\overline{B}S_1\overline{S_0}$$

（3）电路图：

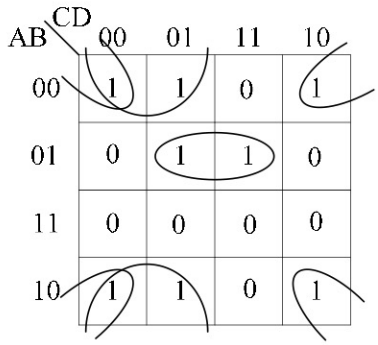


2-9

（1）真值表：

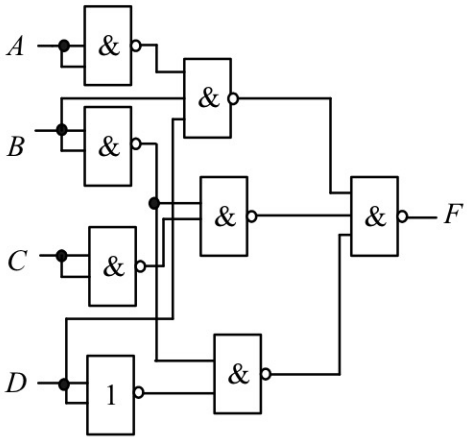
A	B	C	D	F	A	B	C	D	F
0	0	0	0	1	1	0	0	0	1
0	0	0	1	1	1	0	0	1	1
0	0	1	0	1	1	0	1	0	1
0	0	1	1	0	1	0	1	1	0
0	1	0	0	0	1	1	0	0	0
0	1	0	1	1	1	1	0	1	0
0	1	1	0	0	1	1	1	0	0
0	1	1	1	1	1	1	1	1	0

（2）卡诺图及表达式：



$$F=\overline{\overline{B}C}+\overline{\overline{B}D}+\overline{A}BD=\overline{\overline{BC}\,\overline{BD}\,\overline{ABD}}$$

（3）电路图：



2-10

(1) 真值表：

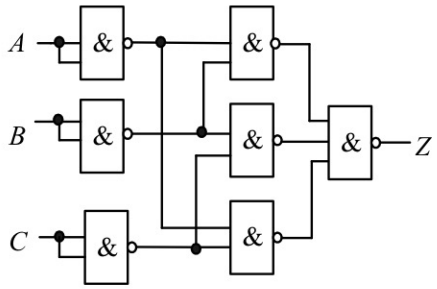
A	B	C	Z
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	0

（2）卡诺图及表达式：

A \ BC	00	01	11	10
0	1	1	0	1
1	1	0	0	0

得 $Z=\overline{\overline{AB}\,\overline{BC}\,\overline{AC}}$

（3）电路图：



2-11

(1)

①真值表：

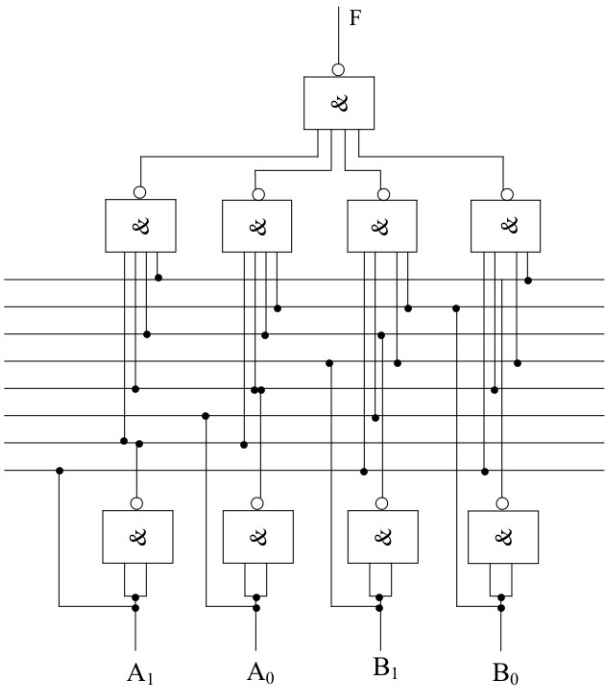
A_1	A_0	B_1	B_0	F	A_1	A_0	B_1	B_0	F
0	0	0	0	1	1	0	0	0	0
0	0	0	1	0	1	0	0	1	0
0	0	1	0	0	1	0	1	0	1
0	0	1	1	0	1	0	1	1	0
0	1	0	0	0	1	1	0	0	0
0	1	0	1	1	1	1	0	1	0
0	1	1	0	0	1	1	1	0	0
0	1	1	1	0	1	1	1	1	1

②卡诺图及表达式：

A_1A_0 \ B_1B_0	00	01	11	10
00	1	0	0	0
01	0	1	0	0
11	0	0	1	0
10	0	0	0	1

$$F=\overline{\overline{A_1A_0B_1B_0}\,\overline{A_1A_0\overline{B_1}B_0}\,\overline{A_1\overline{A_0}B_1B_0}\,\overline{A_1\overline{A_0}\,\overline{B_1}\,\overline{B_0}}}$$

③电路图：



（2）

①真值表：

A_1	A_0	B_1	B_0	F
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	1
0	1	0	0	0
0	1	0	1	0
0	1	1	0	1
0	1	1	1	0

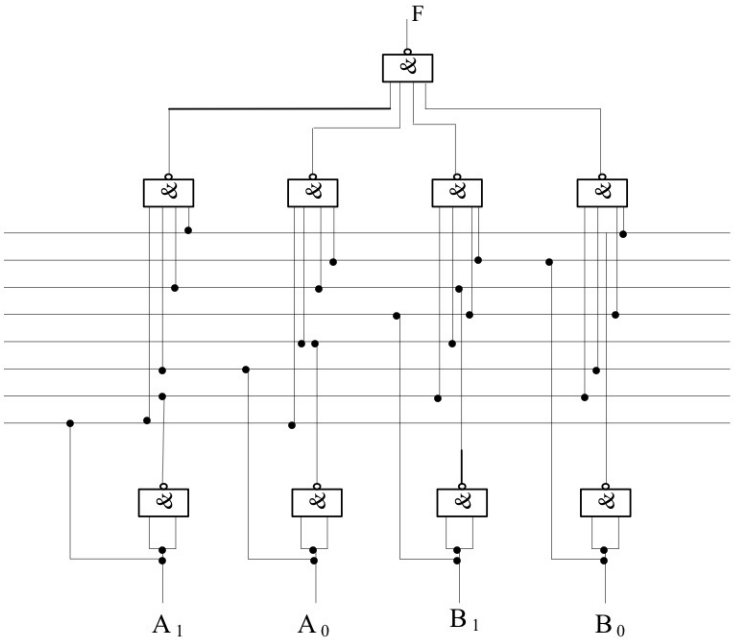
A_1	A_0	B_1	B_0	F
1	0	0	0	0
1	0	0	1	1
1	0	1	0	0
1	0	1	1	0
1	1	0	0	1
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

②卡诺图及表达式：

A_1A_0 \ B_1B_0	00	01	11	10
00	0	0	1	0
01	0	0	0	1
11	1	0	0	0
10	0	1	0	0

$$F=\overline{A_1A_0\overline{B_1}\overline{B_0}\;A_1\overline{A_0}B_1B_0\;\overline{A_1}A_0B_1\overline{B_0}\;\overline{A_1}\overline{A_0}B_1B_0\;\overline{A_1}A_0\overline{B_1}\overline{B_0}}$$

③电路图：



（3）

①真值表：

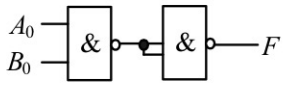
A_1	A_0	B_1	B_0	F	A_1	A_0	B_1	B_0	F
0	0	0	0	0	1	0	0	0	0
0	0	0	1	0	1	0	0	1	0
0	0	1	0	0	1	0	1	0	0
0	0	1	1	0	1	0	1	1	0
0	1	0	0	0	1	1	0	0	0
0	1	0	1	1	1	1	0	1	1
0	1	1	0	0	1	1	1	0	0
0	1	1	1	1	1	1	1	1	1

②卡诺图及表达式：

A_1A_0 \ B_1B_0	00	01	11	10
00	0	0	0	0
01	0	1	1	0
11	0	1	1	0
10	0	0	0	0

$F = \overline{\overline{A_0 B_0}}$

③电路图：



（4）

①真值表：

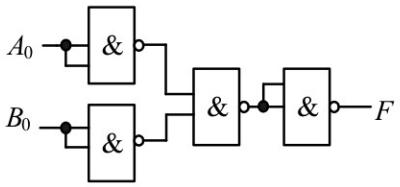
A_1	A_0	B_1	B_0	F	A_1	A_0	B_1	B_0	F
0	0	0	0	1	1	0	0	0	1
0	0	0	1	0	1	0	0	1	0
0	0	1	0	1	1	0	1	0	1
0	0	1	1	0	1	0	1	1	0
0	1	0	0	0	1	1	0	0	0
0	1	0	1	0	1	1	0	1	0
0	1	1	0	0	1	1	1	0	0
0	1	1	1	0	1	1	1	1	0

②卡诺图及表达式：

A_1A_0 \ B_1B_0	00	01	11	10
00	1	0	0	1
01	0	0	0	0
11	0	0	0	0
10	1	0	0	1

$F = \overline{\overline{A_0 B_0}}$

③电路图：



（5）

①真值表：

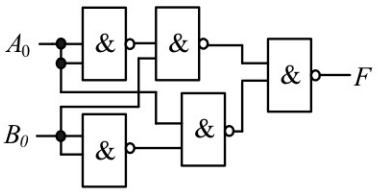
A_1	A_0	B_1	B_0	F	A_1	A_0	B_1	B_0	F
0	0	0	0	0	1	0	0	0	0
0	0	0	1	1	1	0	0	1	1
0	0	1	0	0	1	0	1	0	0
0	0	1	1	1	1	0	1	1	1
0	1	0	0	1	1	1	0	0	1
0	1	0	1	0	1	1	0	1	0
0	1	1	0	1	1	1	1	0	1
0	1	1	1	0	1	1	1	1	0

②卡诺图及表达式：

A_1A_0 \ B_1B_0	00	01	11	10
00	0	1	1	0
01	1	0	0	1
11	1	0	0	1
10	0	1	1	0

$$F = A_0\overline{B_0} + \overline{A_0}B_0 = \overline{\overline{A_0\overline{B_0}}\;\overline{\overline{A_0}B_0}}$$

③电路图：



2-12

（1）真值表：

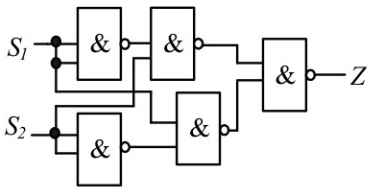
S_1	S_2	Z
0	0	0
0	1	1
1	0	1
1	1	0

（2）卡诺图及表达式：

S_1 \ S_2	0	1
0	0	1
1	1	0

$$Z = \overline{\overline{S_1}S_2S_1\overline{S_2}}$$

（3）电路图：



2-13

（1）真值表：

T_3	T_2	T_1	T_0	X	Y
0	0	0	0	1	0
0	0	0	1	1	0
0	0	1	0	1	0
0	0	1	1	1	0
0	1	0	0	1	0
0	1	0	1	0	0
0	1	1	0	0	0
0	1	1	1	0	0

T_3	T_2	T_1	T_0	X	Y
1	0	0	0	0	0
1	0	0	1	0	0
1	0	1	0	0	0
1	0	1	1	0	1
1	1	0	0	0	1
1	1	0	1	0	1
1	1	1	0	0	1
1	1	1	1	0	1

（2）卡诺图及表达式：

T_3T_2 \ T_1T_0	00	01	11	10
00	1	1	1	1
01	1	0	0	0
11	0	0	0	0
10	0	0	0	0

X 的卡诺图

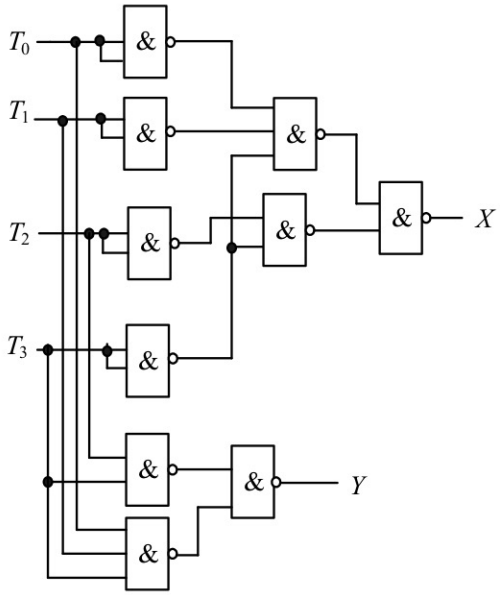
T_3T_2 \ T_1T_0	00	01	11	10
00	0	0	0	0
01	0	0	0	0
11	1	1	1	1
10	0	0	1	0

Y 的卡诺图

$$X=\overline{\overline{T_3T_1T_0}+\overline{T_3T_2}}=\overline{\overline{T_3T_1T_0}\,\overline{T_3T_2}}$$

$$Y=\overline{\overline{T_3T_2}+\overline{T_3T_1T_0}}=\overline{\overline{T_3T_2}\,\overline{T_3T_1T_0}}$$

（3）电路图：



2-14

1) $F_1=\overline{\overline{AB}\,\overline{AC}}$

当$\overline{B}=1,C=1$时存在 0 型冒险

更改设计为

$F_1=\overline{\overline{AB}\,\overline{AC}\,\overline{BC}}$

2） $F_2=\overline{\overline{BD}\,\overline{ABC}}$

当$\overline{D},\overline{C},A=1$时存在 0 型冒险

更改设计为

$F_2 = \overline{\overline{B\bar{D}}\,\overline{AB\bar{C}}\,\overline{A\bar{C}D}}$

3） $F_3 = \overline{\overline{A\bar{C}}\,\overline{B\bar{C}D}}$

当 $\bar{A},\bar{B},D=1$ 存在 0 型冒险。

更改设计为

$F_3 = \overline{\overline{A\bar{C}}\,\overline{B\bar{C}D}\,\overline{AB\bar{D}}}$

4） $F_4 = \overline{\overline{AB\bar{C}}\,\overline{B\bar{C}D}\,\overline{A\bar{C}D}\,\overline{B\bar{C}\bar{D}}}$

当 $\bar{A},\bar{B},\bar{D}=1$ 存在 0 型冒险。

更改设计为

$F_4 = \overline{\overline{AB\bar{C}}\,\overline{B\bar{C}D}\,\overline{A\bar{C}D}\,\overline{B\bar{C}\bar{D}}\,\overline{AB\bar{D}}}$

5） $F_5 = \overline{\overline{B\bar{D}}\,\overline{AB\bar{C}}\,\overline{A\bar{B}C}}$

当 $A=1,C=0,D=0$ 或 $A=0,C=1,D=0$ 存在 0 型冒险。

更改设计为

$F_5 = \overline{\overline{B\bar{D}}\,\overline{AB\bar{C}}\,\overline{A\bar{B}C}\,\overline{A\bar{C}D}\,\overline{A\bar{C}\bar{D}}}$

6） $F_6 = \overline{\overline{AB}\,\overline{B\bar{C}D}\,\overline{B\bar{C}\bar{D}}}$

当 $A=0,C=0,D=0$ 或 $A=0,C=1,D=1$ 存在 0 型冒险。

更改设计为： $F_6 = \overline{\overline{AB}\,\overline{B\bar{C}D}\,\overline{B\bar{C}\bar{D}}\,\overline{A\bar{C}D}\,\overline{A\bar{C}\bar{D}}}$

7） $F_7 = \overline{\overline{B\bar{C}\bar{D}}\,\overline{A\bar{B}D}\,\overline{A\bar{C}}}$

当 $A=0,B=1,D=0$ 或 $B=0,C=1,D=1$ 存在 0 型冒险。

更改设计为： $F_7 = \overline{\overline{B\bar{C}D}\,\overline{A\bar{B}D}\,\overline{A\bar{C}}\,\overline{AB\bar{D}}\,\overline{B\bar{C}\bar{D}}}$

2-15

a) $Z_1 = \overline{\overline{A\bar{B}D}\,\overline{B\bar{C}\bar{D}}} = A\bar{B}D + B\bar{C}\bar{D}$

当 $A=1,C=0,D=1$ 时存在 0 型冒险。

当 $A=0,C=1$ 存在 1 型冒险

当 $B=1,D=0$ 时，存在 1 型冒险

b) $Z_2 = \overline{\overline{A + B + \overline{C}} + \overline{B + C + D} + \overline{\overline{A} + \overline{C} + \overline{D}}} = \left(A + B + \overline{C}\right)\left(B + C + D\right)\left(\overline{A} + \overline{C} + \overline{D}\right)$

当 $C = 1, D = 0$ 时，存在 1 型冒险

当 $B = 1, D = 0$ 时，存在 1 型冒险

当 $A = 0, C = 1$ 时，存在 1 冒险

第三章习题

3-1

（a）$\overline{Y}_{EX}$、$\overline{Y}_2$、$\overline{Y}_1$、$\overline{Y}_0$、Y_S的值分别为：0、0、0、1、1

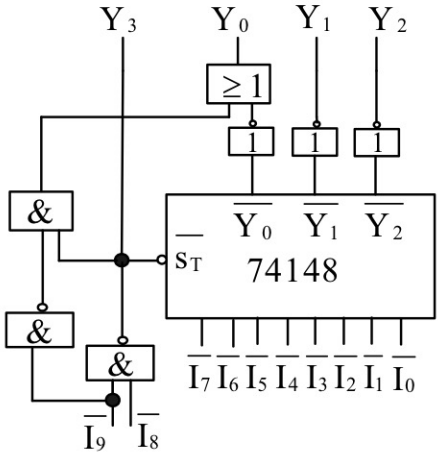
（b）$\overline{Y}_{EX}$、$\overline{Y}_2$、$\overline{Y}_1$、$\overline{Y}_0$、Y_S的值分别为：0、0、1、1、1

3-2

（1）设计真值表：

$\overline{I}_9$	$\overline{I}_8$	$\overline{I}_7$	$\overline{I}_6$	$\overline{I}_5$	$\overline{I}_4$	$\overline{I}_3$	$\overline{I}_2$	$\overline{I}_1$	$\overline{I}_0$	Y_3	Y_2	Y_1	Y_0
0	×	×	×	×	×	×	×	×	×	1	0	0	1
1	0	×	×	×	×	×	×	×	×	1	0	0	0
1	1	0	×	×	×	×	×	×	×	0	1	1	1
1	1	1	0	×	×	×	×	×	×	0	1	1	0
1	1	1	1	0	×	×	×	×	×	0	1	0	1
1	1	1	1	1	0	×	×	×	×	0	1	0	0
1	1	1	1	1	1	0	×	×	×	0	0	1	1
1	1	1	1	1	1	1	0	×	×	0	0	1	0
1	1	1	1	1	1	1	1	0	×	0	0	0	1
1	1	1	1	1	1	1	1	1	0	0	0	0	0

（2）电路图：



3-3

（1）0111

（2）0110

（3）0111

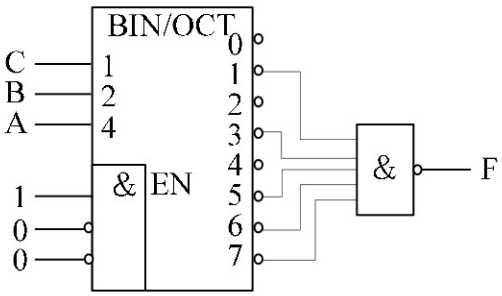
（4）0110

（5）0110

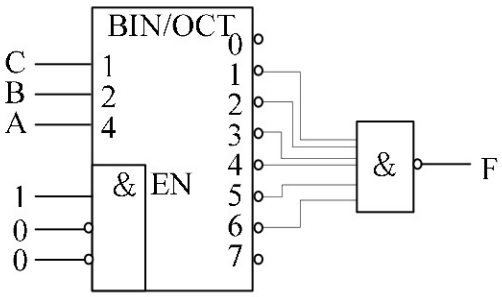
（6）0110

3-4

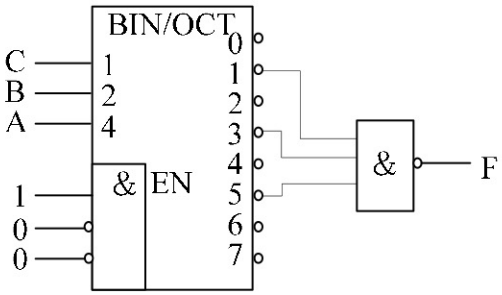
（1）$F_1(A,B,C)=AB+C$



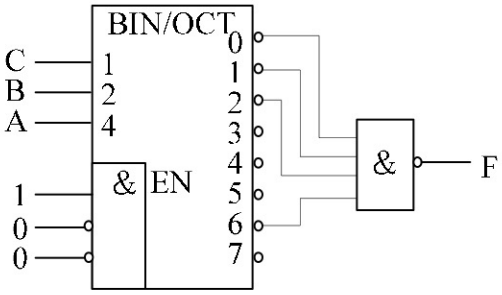
（2）$F_2(A,B,C)=A\overline{B}+B\overline{C}+C\overline{A}$



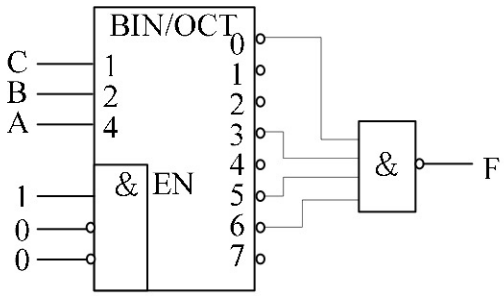
（3）$F_3(A,B,C)=(\overline{A}+\overline{B})C$



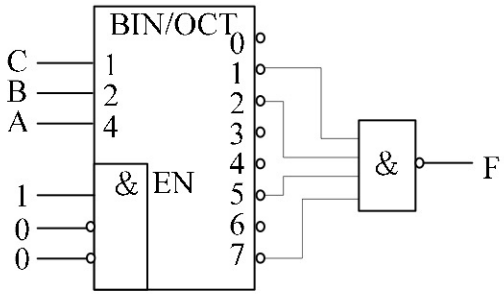
（4）$F_4(A,B,C)=(\overline{A}+B)(\overline{B}+\overline{C})$



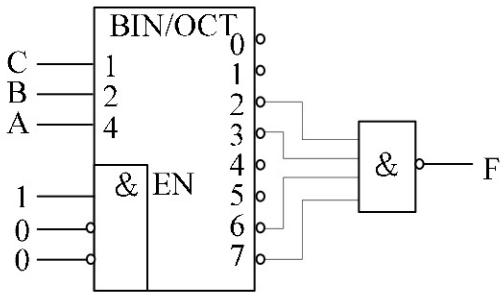
（5）$F_5(A,B,C)=\sum m(0,3,5,6)$



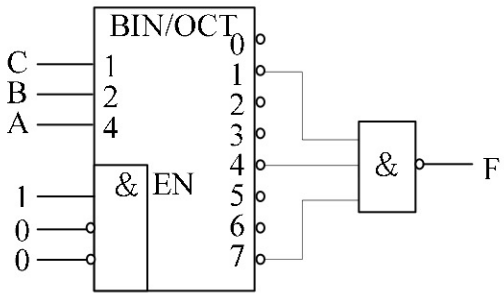
（6） $F_6(A,B,C)=\sum m(1,2,5,7)$



（7） $F_7(A,B,C)=\prod M(0,1,4,5)$

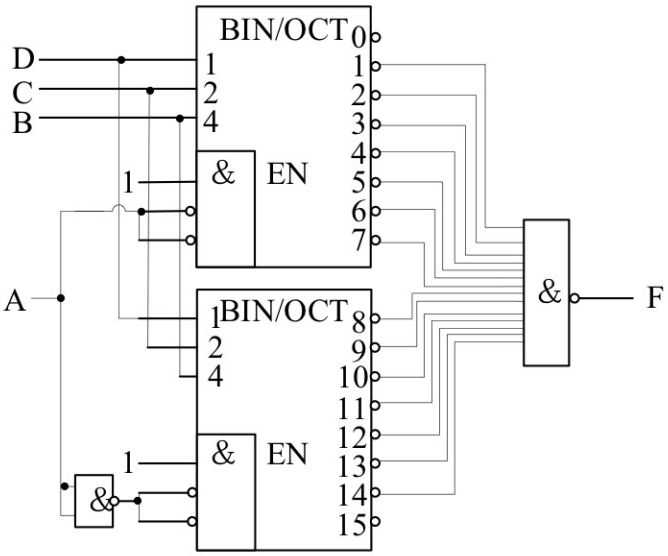


（8） $F_8(A,B,C)=\prod M(0,2,3,5,6)$

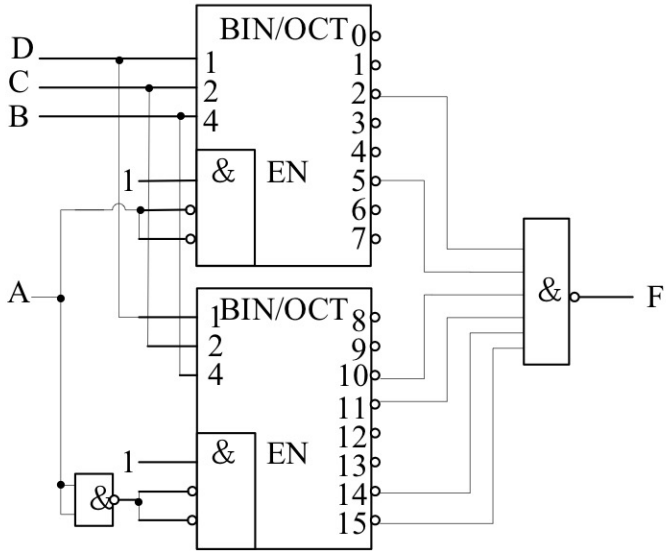


3-5

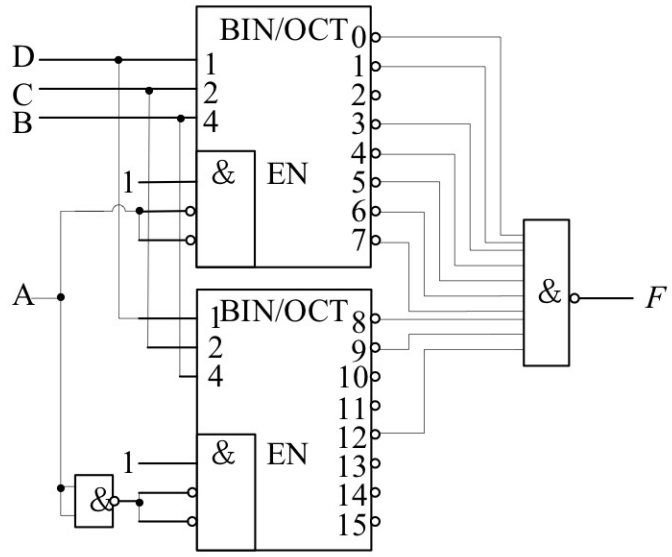
（1） $F_1(A,B,C,D)=A\overline{B}+B\overline{C}+C\overline{D}+D\overline{A}$



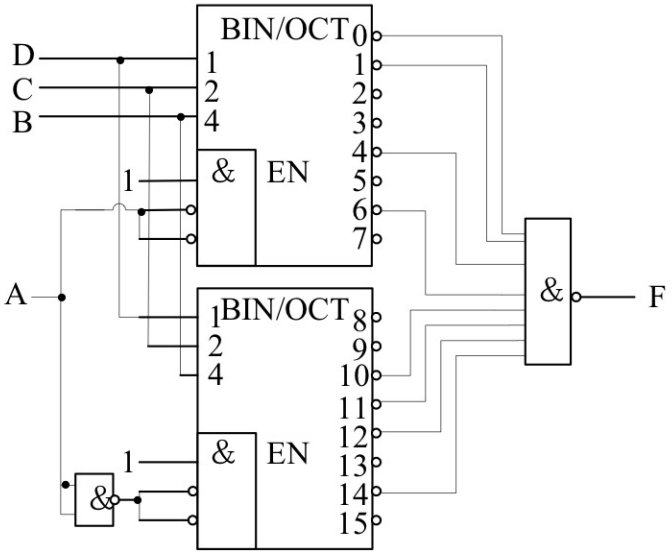
（2） $F_2(A,B,C,D) = AC + \overline{B}C\overline{D} + \overline{A}B\overline{C}D$



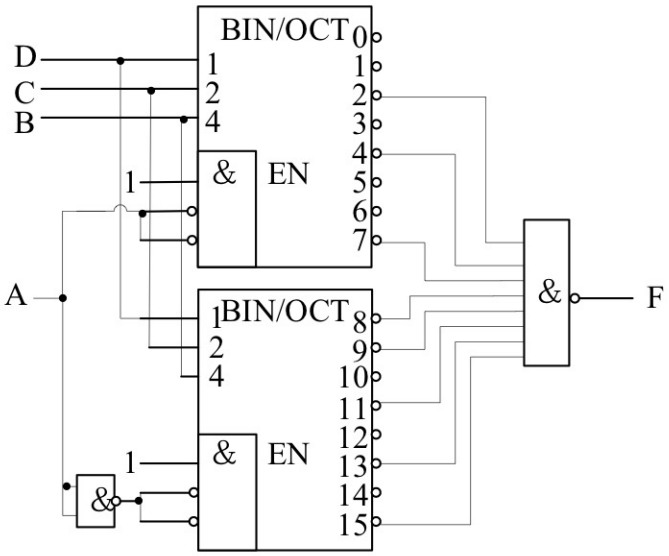
（3） $F_3(A,B,C,D) = \overline{A}(B + D) + \overline{C + BD}$



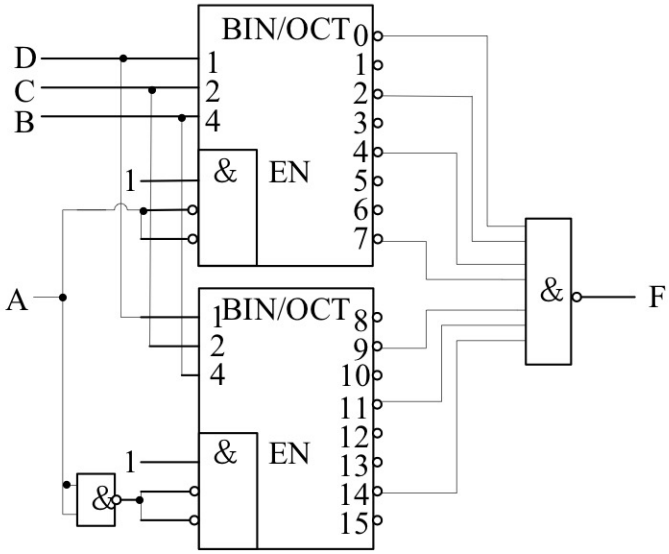
（4） $F_4(A,B,C,D)=\sum m(0,1,4,6,10,11,12,14)$



（5） $F_5(A,B,C,D)=\sum m(2,4,7,8,9,11,13,15)$



（6） $F_6(A,B,C,D)=\sum M(1,3,5,6,8,10,12,13,15)$



3-6

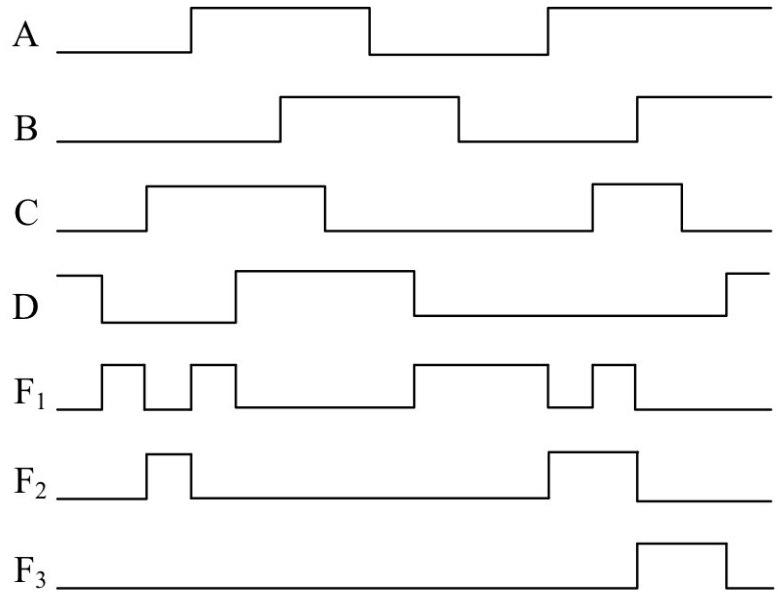
$$\overline{Y}_0 = A_3 + A_2 + A_1 + A_0$$
$$\overline{Y}_1 = A_3 + A_2 + A_1 + \overline{A}_0$$
$$\overline{Y}_2 = A_3 + A_2 + \overline{A}_1 + A_0$$
$$\overline{Y}_3 = A_3 + A_2 + \overline{A}_1 + \overline{A}_0$$
$$\overline{Y}_4 = A_3 + \overline{A}_2 + A_1 + A_0$$
$$\overline{Y}_5 = A_3 + \overline{A}_2 + A_1 + \overline{A}_0$$
$$\overline{Y}_6 = A_3 + \overline{A}_2 + \overline{A}_1 + A_0$$
$$\overline{Y}_7 = A_3 + \overline{A}_2 + \overline{A}_1 + \overline{A}_0$$
$$\overline{Y}_8 = \overline{A}_3 + A_2 + A_1 + A_0$$
$$\overline{Y}_9 = \overline{A}_3 + A_2 + A_1 + \overline{A}_0$$

3-7

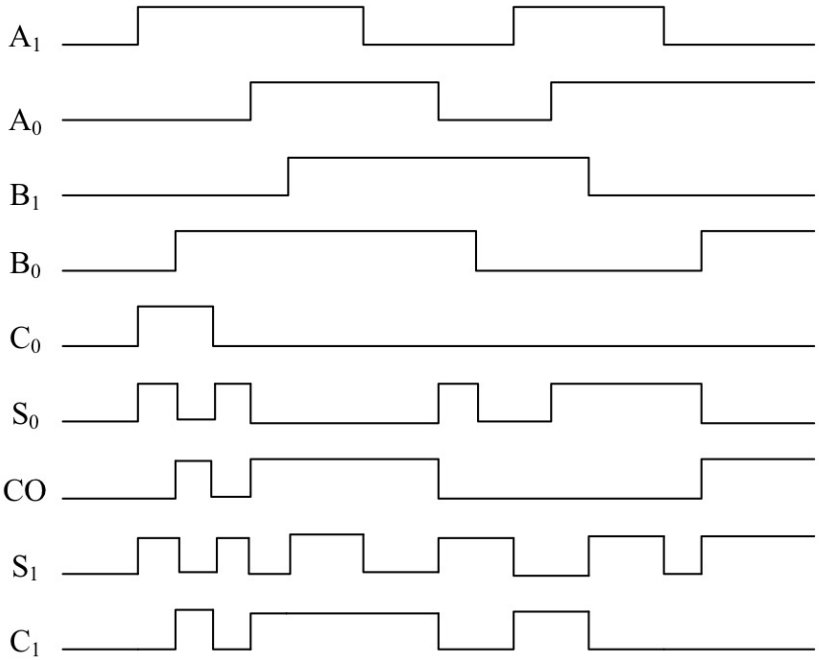
$$F_1 = \overline{(D+C+B+A)(D+C+\overline{B}+A)(D+\overline{C}+B+\overline{A})} = \overline{A}\overline{C}\overline{D} + A\overline{B}C\overline{D}$$

$$F_2 = \overline{(D+\overline{C}+B+A)(D+C+B+\overline{A})(D+\overline{C}+\overline{B}+A)} = A\overline{B}\overline{C}\overline{D} + \overline{A}C\overline{D}$$

$$F_3 = \overline{(D+C+\overline{B}+\overline{A})(D+\overline{C}+\overline{B}+\overline{A})(\overline{D}+C+B+\overline{A})} = AB\overline{D} + A\overline{B}\overline{C}D$$



3-8



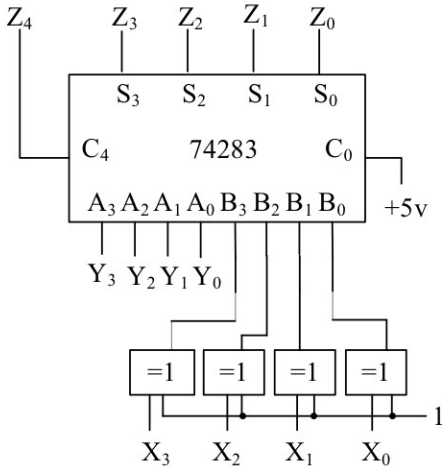
3-9

$$F_1 = \overline{(A?\ B)ABCD}$$

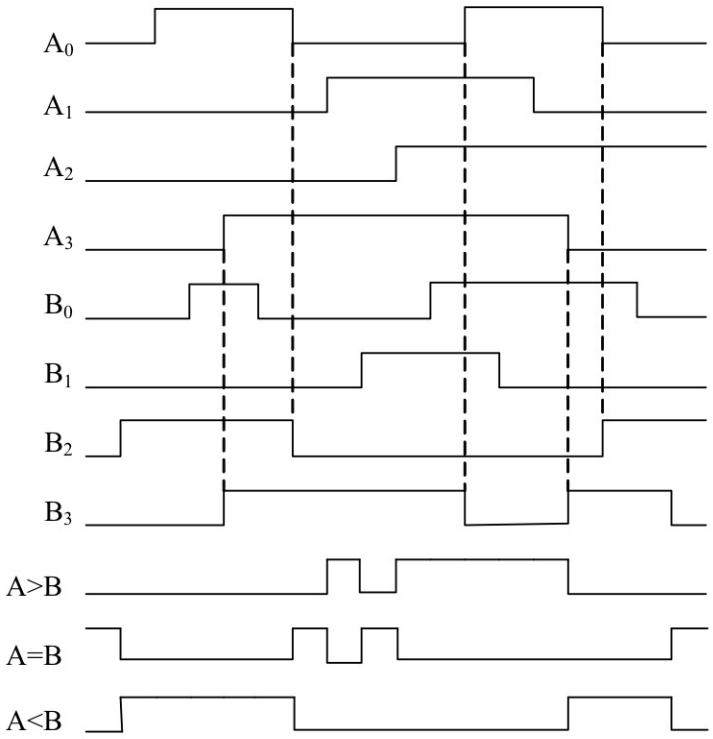
$$F_2 = \overline{(C\text{排}D)(A\quad B)C}$$

3-10

求减数的补码，然后与被减数相加即可。电路图如下：

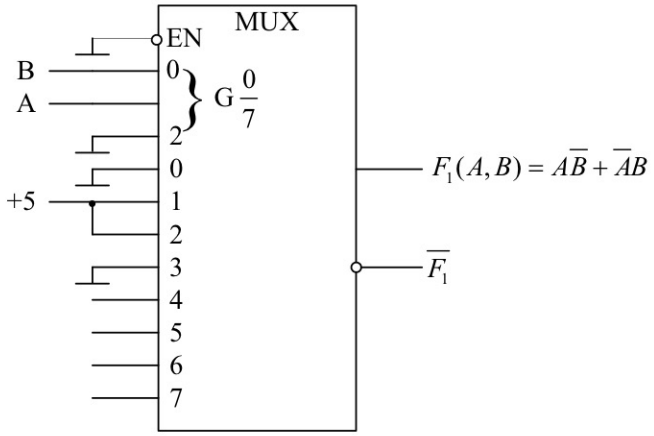


3-11

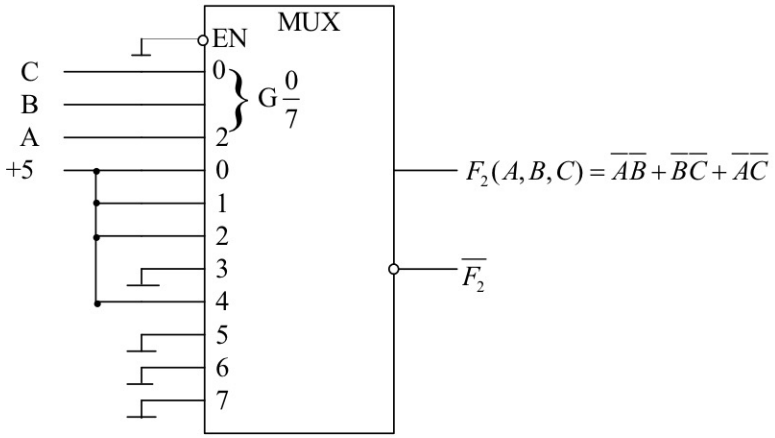


3-12

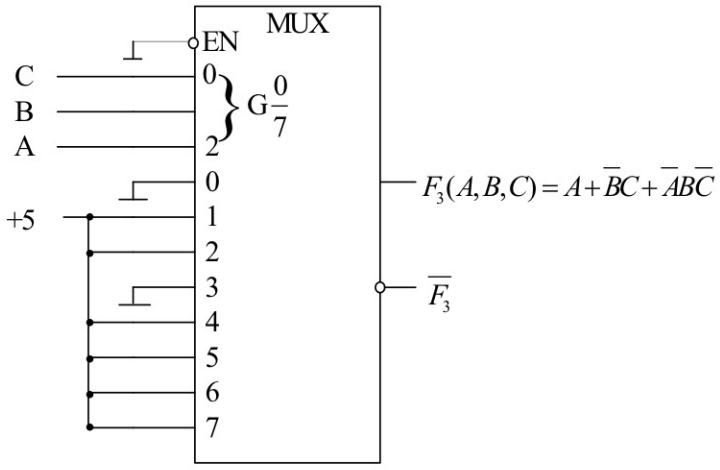
（1）


MUX
EN
B
A
+5
0
G 0/7
2
0
1
2
3
4
5
6
7
$F_1(A,B)=A\overline{B}+\overline{A}B$
$\overline{F_1}$

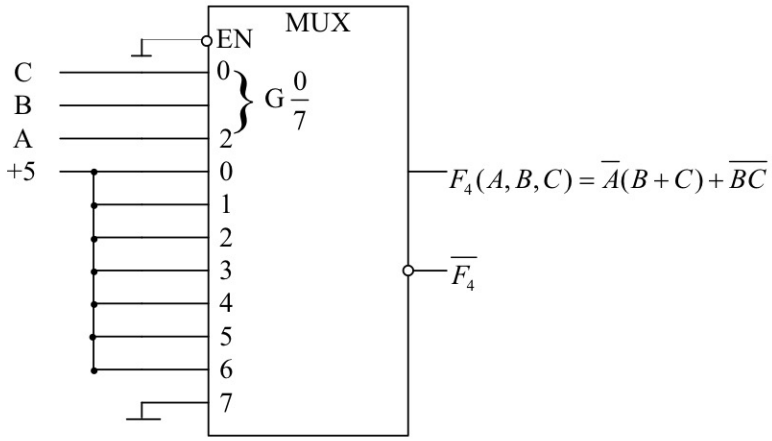
（2）


MUX
EN
C
B
A
+5
0
G 0/7
2
0
1
2
3
4
5
6
7
$F_2(A,B,C)=\overline{A}\overline{B}+\overline{B}\overline{C}+\overline{A}\overline{C}$
$\overline{F_2}$

（3）

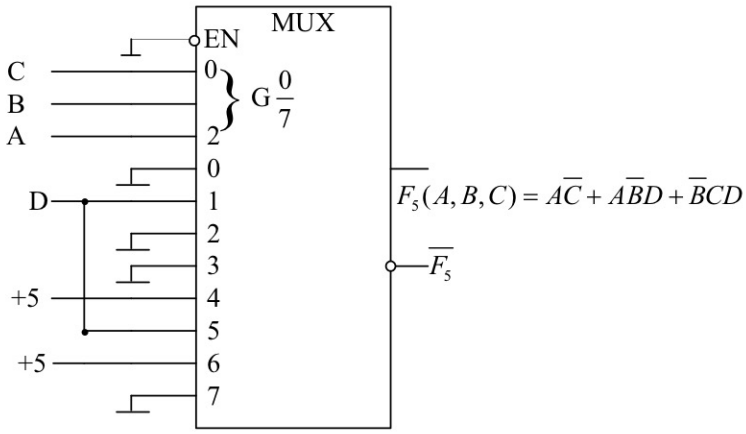

MUX
EN
C
B
A
+5
0
G 0/7
2
0
1
2
3
4
5
6
7
$F_3(A,B,C)=A+\overline{B}C+\overline{A}B\overline{C}$
$\overline{F_3}$

（4）

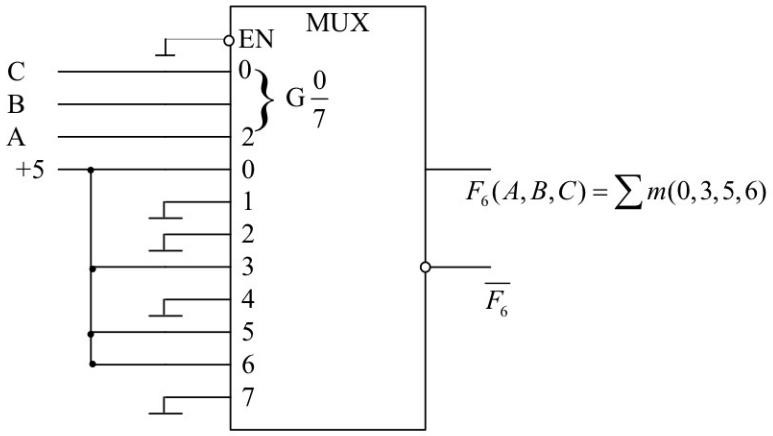


（5）

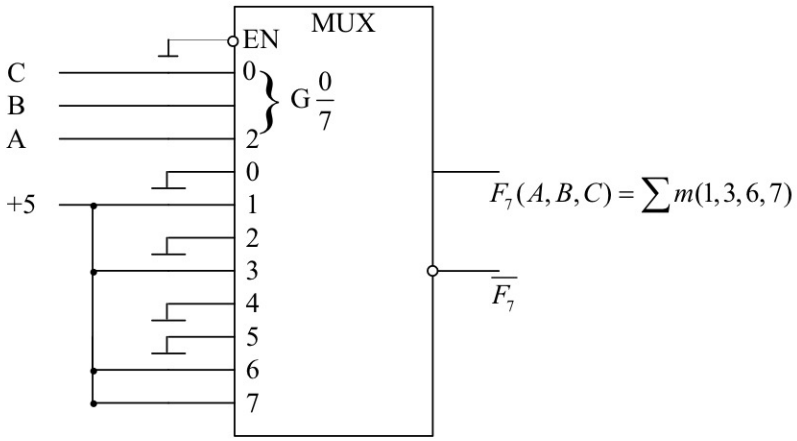
A	B	C	D	F	A	B	C	D	F
0	0	0	0	0	1	0	0	0	1
0	0	0	1	0	1	0	0	1	1
0	0	1	0	0	1	0	1	0	0
0	0	1	1	1	1	0	1	1	1
0	1	0	0	0	1	1	0	0	1
0	1	0	1	0	1	1	0	1	1
0	1	1	0	0	1	1	1	0	0
0	1	1	1	0	1	1	1	1	0



（6）

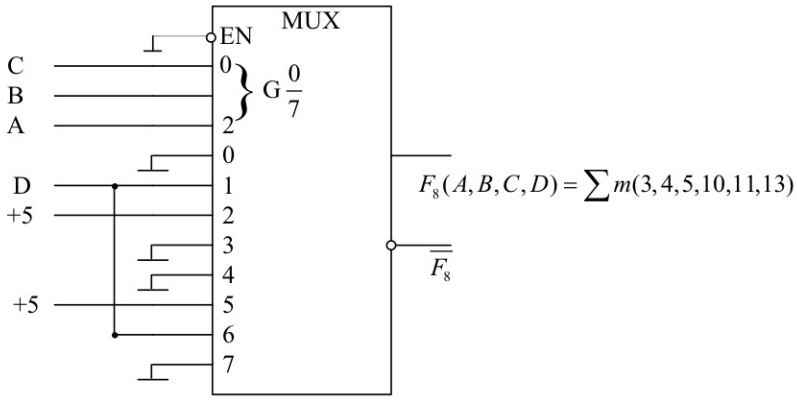


（7）



（8）

A	B	C	D	F	A	B	C	D	F
0	0	0	0	0	1	0	0	0	0
0	0	0	1	0	1	0	0	1	0
0	0	1	0	0	1	0	1	0	1
0	0	1	1	1	1	0	1	1	1
0	1	0	0	1	1	1	0	0	0
0	1	0	1	1	1	1	0	1	1
0	1	1	0	0	1	1	1	0	0
0	1	1	1	0	1	1	1	1	0

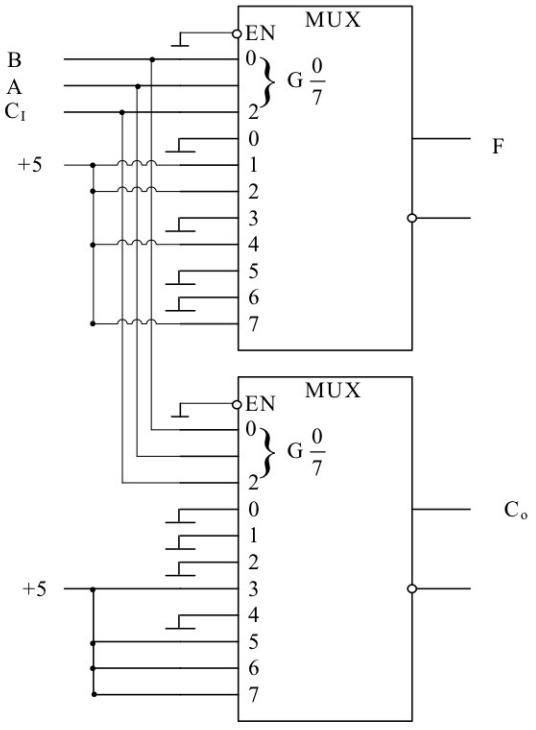


3-13

(1)真值表：

C_I	A	B	F	C_O
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

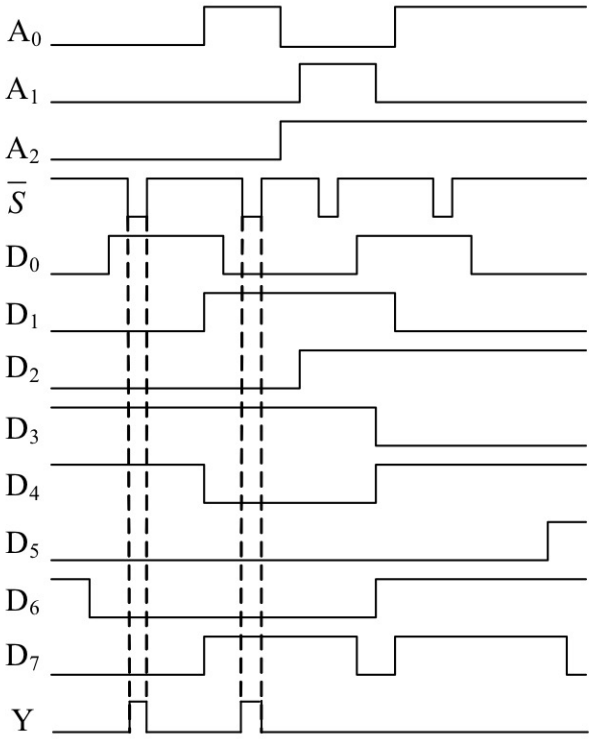
（2）电路图



3-14

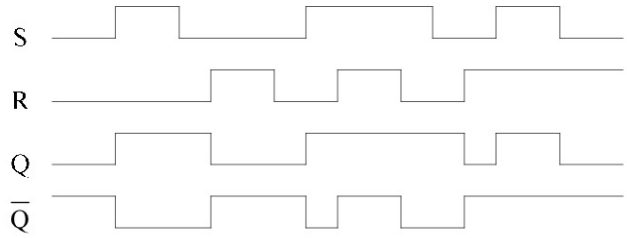
$$Y = \overline{A}\overline{B}D_0 + \overline{A}BD_1 + A\overline{B}D_2 + ABD_3 = \overline{A}\overline{B} + \overline{A}B\overline{CD} + AB\overline{D}$$

3-15

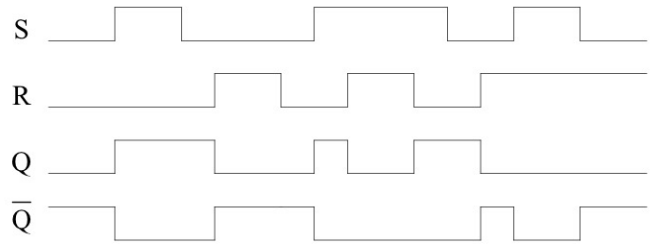


第四章习题

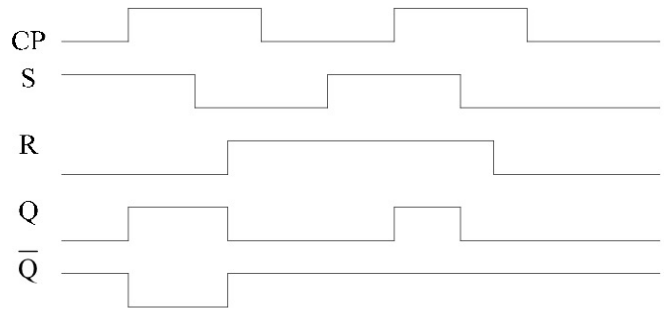
4-1



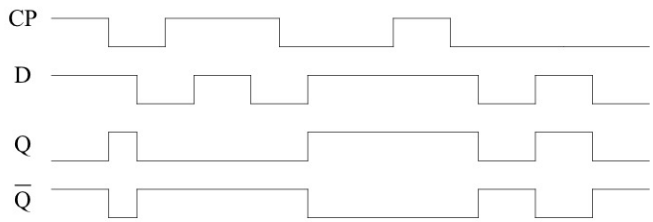
4-2



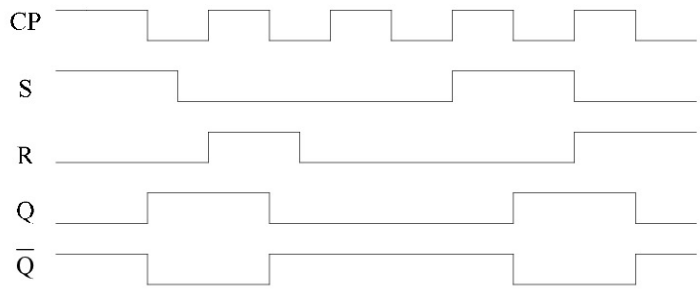
4-3



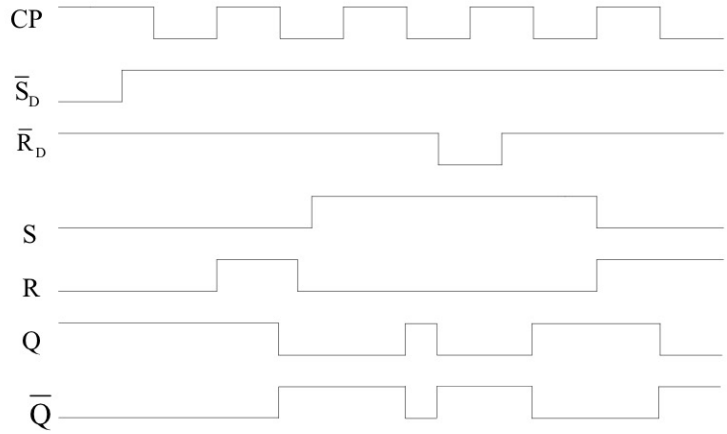
4-4



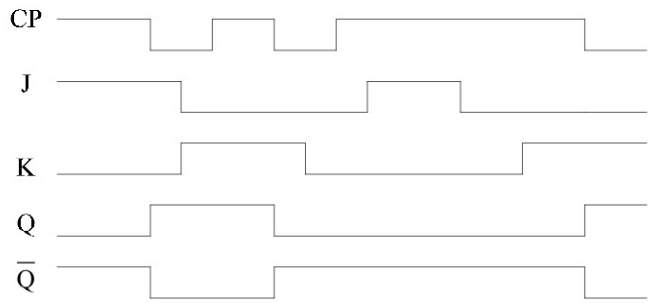
4-5


CP
S
R
Q
$\overline{Q}$

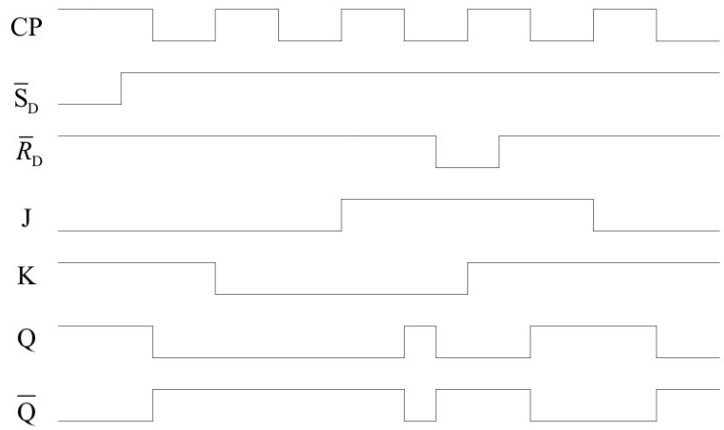
4-6


CP
$\overline{S}_D$
$\overline{R}_D$
S
R
Q
$\overline{Q}$

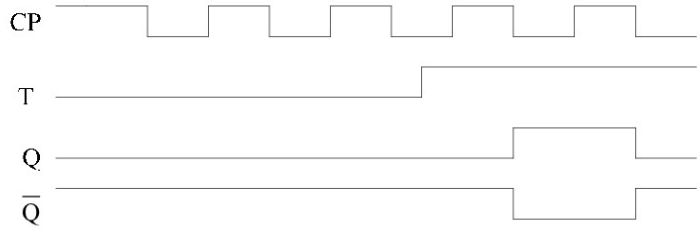
4-7


CP
J
K
Q
$\overline{Q}$

4-8

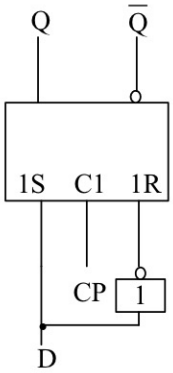

CP
$\overline{S}_D$
$\overline{R}_D$
J
K
Q
$\overline{Q}$

4-9

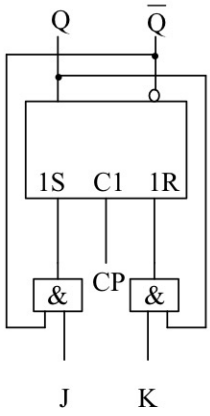


4-10

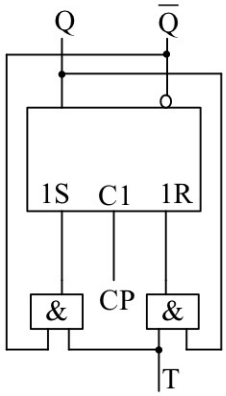
RS → D



RS → JK



RS → T



4-11

(1)转换真值表

T → RS

R	S	Q^n	Q^{n+1}	T
0	0	0	0	0
0	0	1	1	0
0	1	0	1	1
0	1	1	1	0
1	0	0	0	0
1	0	1	0	1
1	1	0	×	×
1	1	1	×	×

$T = S\overline{Q}^n + RQ^n$

T → JK

J	K	Q^n	Q^{n+1}	T
0	0	0	0	0
0	0	1	1	0
0	1	0	0	0
0	1	1	0	1
1	0	0	1	1
1	0	1	1	0
1	1	0	1	1
1	1	1	0	1

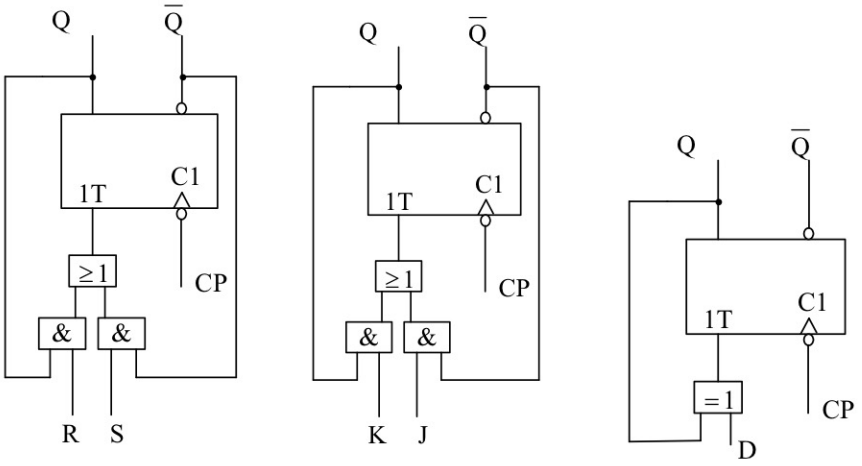
$T = J\overline{Q}^n + KQ^n$

T → D

D	Q^n	Q^{n+1}	T
0	0	0	0
0	1	0	1
1	0	1	1
1	1	1	0

$$T = D \oplus Q^n$$

(2)转换电路图



4-12

（1）$Q_0^{n+1} = D = \bar{Q}_0^n, CP\downarrow$

（2）$Q_1^{n+1} = J\bar{Q}_1^n + \bar{K}Q_1^n = A\bar{Q}_1^n + AQ_1^n = A, CP\downarrow$

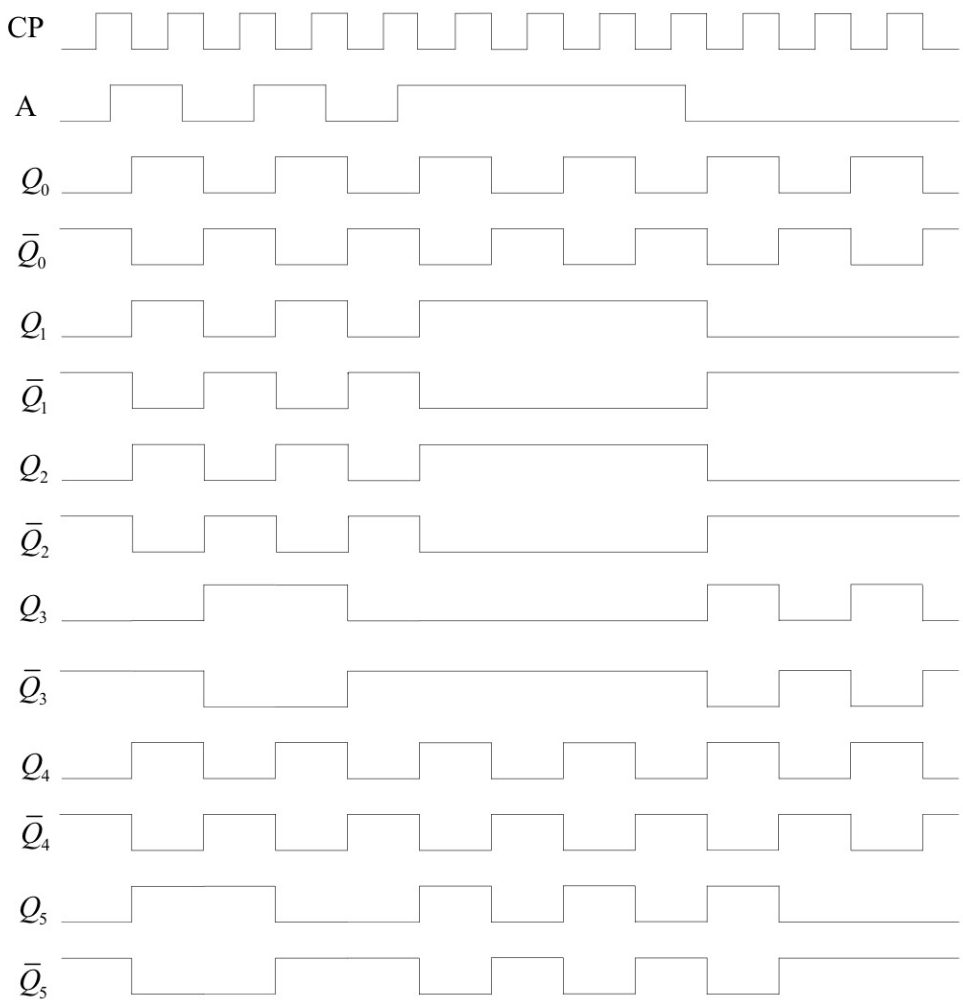
（3）$Q_2^{n+1} = T \oplus Q_2^n = A \oplus Q_2^n \oplus Q_2^n = A, CP\downarrow$

（4）$Q_3^{n+1} = D = A \oplus \bar{Q}_3^n = A \odot Q_3^n, CP\downarrow$

（5）$Q_4^{n+1} = J\bar{Q}_4^n + \bar{K}Q_4^n = \bar{Q}_4^n, CP\downarrow$

（6）$Q_5^{n+1} = J\bar{Q}_5^n + \bar{K}Q_5^n = (A + Q_5^n)\bar{Q}_5^n + (\overline{A + \bar{Q}_5^n})Q_5^n, CP\downarrow$

$= A \oplus Q_5^n$



4-13

(1)列写方程

时钟方程 $CP_0 = CP_1 = CP_2 = CP$

驱动方程 $J_0 = 1 \quad J_1 = \overline{Q}_0^n \quad J_2 = \overline{Q}_0^n \overline{Q}_1^n$

$K_0 = 1 \quad K_1 = \overline{Q}_0^n \quad K_2 = \overline{Q}_0^n \overline{Q}_1^n$

特性方程 $Q^{n+1} = J\overline{Q}^n + \overline{K}Q^n$

状态方程 $Q_0^{n+1} = \overline{Q}_0^n$

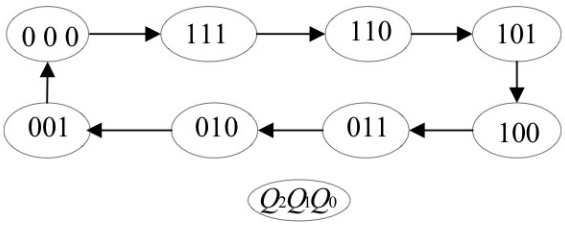
$Q_1^{n+1} = \overline{Q}_0^n \overline{Q}_1^n + Q_0^n Q_1^n = Q_0^n \odot Q_1^n$

$Q_2^{n+1} = \overline{Q}_0^n \overline{Q}_1^n \overline{Q}_2^n + \overline{\overline{Q}_0^n \overline{Q}_1^n} Q_2^n = \left(\overline{Q}_0^n \overline{Q}_1^n\right)? \ Q_2^n$

（2）状态转换表

Q_2^n	Q_1^n	Q_0^n	Q_2^{n+1}	Q_1^{n+1}	Q_0^{n+1}
0	0	0	1	1	1
0	0	1	0	0	0
0	1	0	0	0	1
0	1	1	0	1	0
1	0	0	0	1	1
1	0	1	1	0	0
1	1	0	1	0	1
1	1	1	1	1	0

（3）状态图



4-14

(1) 列写方程

时钟方程 $CP_0 = CP_1 = CP_2 = CP$

驱动方程 $T_0 = Q_2^n, T_1 = Q_0^n, T_2 = Q_1^n$

特性方程 $Q^{n+1} = T\overline{Q}^n + \overline{T}Q^n$

$$Q_0^{n+1} = Q_2^n \ ? \ Q_0^n$$

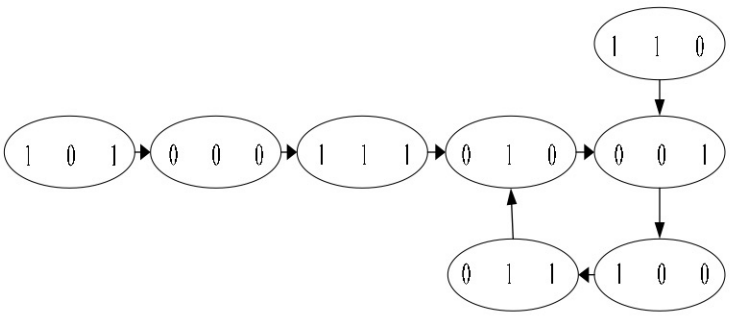
状态方程 $Q_1^{n+1} = Q_0^n \ ? \ Q_1^n$

$$Q_2^{n+1} = Q_1^n \ ? \ Q_2^n$$

(2) 状态转换表

Q_2^n	Q_1^n	Q_0^n	Q_2^{n+1}	Q_1^{n+1}	Q_0^{n+1}
0	0	0	0	0	0
0	0	1	0	1	1
0	1	0	1	1	0
0	1	1	1	0	1
1	0	0	1	0	1
1	0	1	1	1	0
1	1	0	0	1	1
1	1	1	0	0	0

(3) 状态图



4-15

（1）列写方程

$CP_0 = CP$?

时钟方程 $CP_1 = Q_0$?

$CP_2 = \overline{Q}_1$ 碘 Q_1

驱动方程 $T_0 = T_1 = T_2 = 1$

$Q_0^{n+1} = \overline{Q}_0^n, CP$?

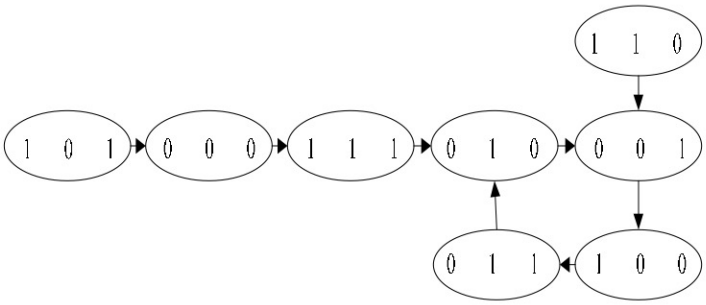
状态方程 $Q_1^{n+1} = \overline{Q}_1^n, Q_0$?

$Q_2^{n+1} = \overline{Q}_2^n, Q_1$?

（2）状态转换表

Q_2^n	Q_1^n	Q_0^n	Q_2^{n+1}	Q_1^{n+1}	Q_0^{n+1}	CP	CP_0	CP_1	CP_2
0	0	0	0	0	1	↓	↓		
0	0	1	1	1	0	↓	↓	↓	↓
0	1	0	0	1	1	↓	↓		
0	1	1	0	0	0	↓	↓	↓	
1	0	0	1	0	1	↓	↓		
1	0	1	0	1	0	↓	↓	↓	↓
1	1	0	1	1	1	↓	↓		
1	1	1	1	0	0	↓	↓	↓	

（3）状态图



4-16

(1) 列写方程

时钟方程 $CP_0 = CP$?

$CP_1 = Q_0$?

$CP_2 = Q_0$?

驱动方程 $J_0 = K_0 = 1$

$J_1 = K_1 = \overline{Q}_2^n$

$J_2 = K_2 = Q_1^n$

状态方程 $Q_0^{n+1} = \overline{Q}_0^n, CP\downarrow$

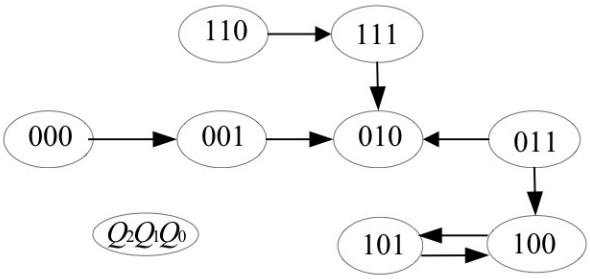
$Q_1^{n+1} = \overline{Q}_2^n \overline{Q}_1^n + Q_2^n Q_1^n, Q_0\downarrow$

$Q_1^{n+1} = \overline{Q}_2^n Q_1^n + Q_2^n \overline{Q}_1^n, Q_0\downarrow$

(2) 状态转换表

Q_2^n	Q_1^n	Q_0^n	Q_2^{n+1}	Q_1^{n+1}	Q_0^{n+1}	CP	CP_0	CP_1	CP_2
0	0	0	0	0	1	↓	↓		
0	0	1	0	1	0	↓	↓	↓	↓
0	1	0	0	1	1	↓	↓		
0	1	1	1	0	0	↓	↓	↓	↓
1	0	0	1	0	1	↓	↓		
1	0	1	1	0	0	↓	↓	↓	↓
1	1	0	1	1	1	↓	↓		
1	1	1	0	1	0	↓	↓	↓	↓

（3）状态图

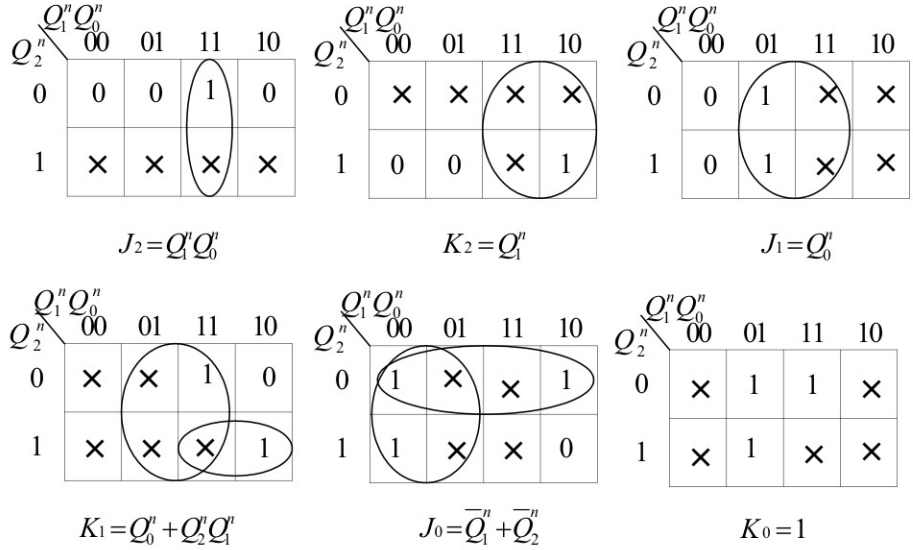


4-17

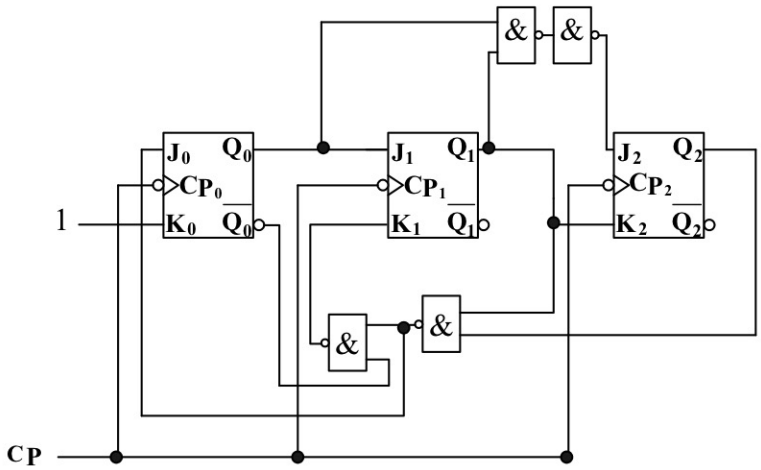
(1) 状态转换表

Q_2^n	Q_1^n	Q_0^n	Q_2^{n+1}	Q_1^{n+1}	Q_0^{n+1}	J_2	K_2	J_1	K_1	J_0	K_0
0	0	0	0	0	1	0	×	0	×	1	×
0	0	1	0	1	0	0	×	1	×	×	1
0	1	0	0	1	1	0	×	×	0	1	×
0	1	1	1	0	0	1	×	×	1	×	1
1	0	0	1	0	1	×	0	0	×	1	×
1	0	1	1	1	0	×	0	1	×	×	1
1	1	0	0	0	0	×	1	×	1	0	×
1	1	1	×	×	×	×	×	×	×	×	×

（2）卡诺图及驱动方程



（3）电路图



4-18

（1） 时钟方程$CP_0=CP_1=CP_2=CP$

（2） 状态转换表

Q_2^n	Q_1^n	Q_0^n	Q_2^{n+1}	Q_1^{n+1}	Q_0^{n+1}	D_2	D_1	D_0
0	0	0	0	0	1	0	0	1
0	0	1	0	1	0	0	1	0
0	1	0	0	1	1	0	1	1
0	1	1	1	0	0	1	0	0
1	0	0	1	1	1	1	1	1
1	0	1	0	0	0	0	0	0
1	1	0	1	0	1	1	0	1
1	1	1	1	1	0	1	1	0

（3）卡诺图及驱动方程

Q_2^n \ $Q_1^nQ_0^n$	00	01	11	10
0	0	0	1	0
1	1	0	1	1

$$D_2 = Q_2^n\overline{Q}_0^n + Q_1^nQ_0^n$$

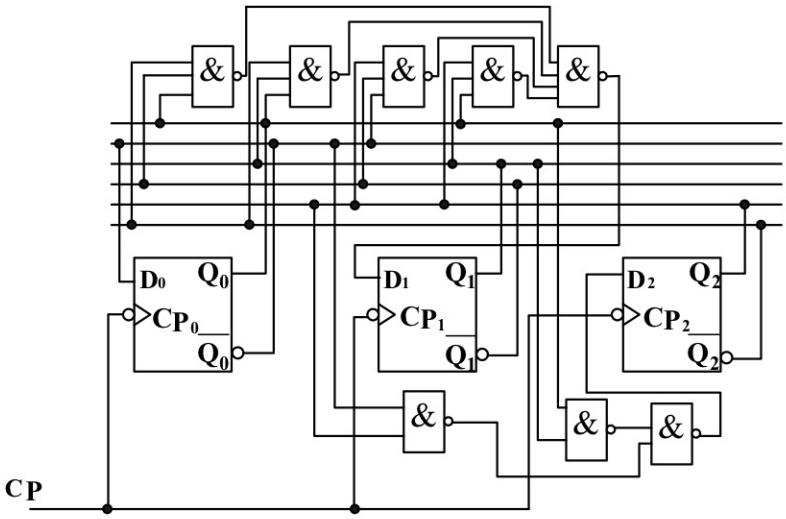
Q_2^n \ $Q_1^nQ_0^n$	00	01	11	10
0	0	1	0	1
1	1	0	1	0

$$D_1 = \overline{\overline{Q}_2^n\overline{Q}_1^nQ_0^n\,\overline{Q}_2^nQ_1^n\overline{Q}_0^n\,\overline{Q}_2^nQ_0^n\,\overline{Q}_1^n\overline{Q}_0^n\,\overline{Q_2^nQ_1^nQ_0^n}}$$

Q_2^n \ $Q_1^nQ_0^n$	00	01	11	10
0	1	0	0	1
1	1	0	0	1

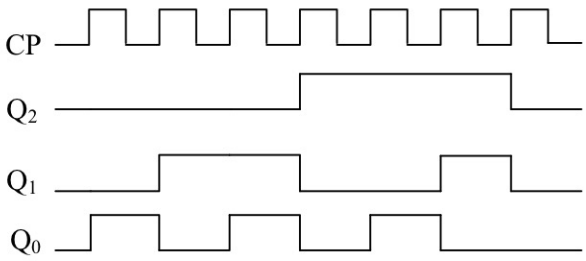
$$D_0 = \overline{Q}_0^n$$

（4）电路图



4-19

（1） 时序图与时钟方程：



$CP_0 = CP$?

$CP_1 = CP$?

$CP_2 = \overline{Q}_1$?

（2）状态转换表

Q_2^n	Q_1^n	Q_0^n	Q_2^{n+1}	Q_1^{n+1}	Q_0^{n+1}	J_2	K_2	J_1	K_1	J_0	K_0	CP_2	CP_1	CP_0
0	0	0	0	0	1	×	×	0	×	1	×		↑	↑
0	0	1	0	1	0	×	×	1	×	×	1		↑	↑
0	1	0	0	1	1	×	×	×	0	1	×		↑	↑
0	1	1	1	0	0	1	×	×	1	×	1	↑	↑	↑
1	0	0	1	0	1	×	×	0	×	1	×		↑	↑
1	0	1	1	1	0	×	×	1	×	×	1		↑	↑
1	1	0	0	0	0	×	1	×	1	0	×	↑	↑	↑
1	1	1	×	×	×	×	×	×	×	×	×		↑	↑

（3）卡诺图及驱动方程

Q_2^n \ $Q_1^n Q_0^n$	00	01	11	10
0	×	×	1	×
1	×	×	×	×

$J_2=1$

Q_2^n \ $Q_1^n Q_0^n$	00	01	11	10
0	×	×	×	×
1	×	×	×	1

$K_2=1$

Q_2^n \ $Q_1^n Q_0^n$	00	01	11	10
0	0	1	×	×
1	0	1	×	×

$J_1 = Q_0^n$

Q_2^n \ $Q_1^n Q_0^n$	00	01	11	10
0	×	×	1	0
1	×	×	×	1

$K_1 = Q_2^n + Q_0^n$

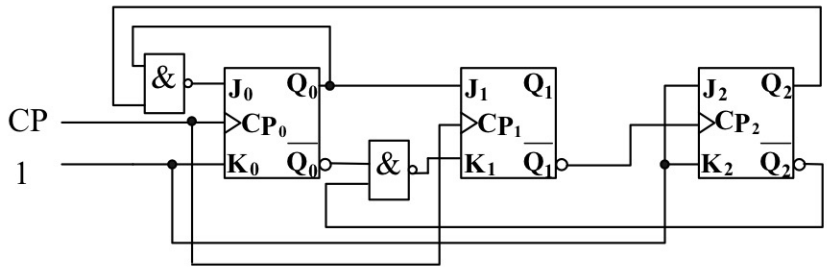
Q_2^n \ $Q_1^n Q_0^n$	00	01	11	10
0	1	×	×	1
1	1	×	×	0

$J_0 = \overline{Q}_2^n + \overline{Q}_0^n$

Q_2^n \ $Q_1^n Q_0^n$	00	01	11	10
0	×	1	1	×
1	×	1	×	×

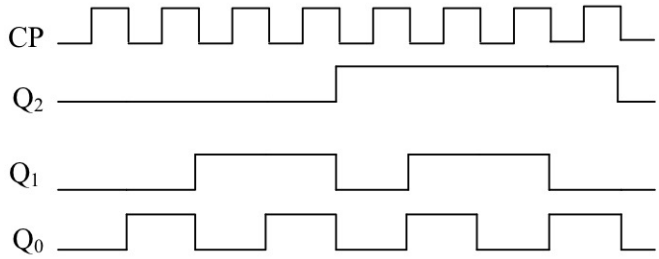
$K_0 = 1$

（4）电路图



4-20

（1） 时序图与时钟方程



$CP_0 = CP$?

$CP_1 = CP$?

$CP_2 = Q_0$?

（2） 状态转换表

Q_2^n	Q_1^n	Q_0^n	Q_2^{n+1}	Q_1^{n+1}	Q_0^{n+1}	D_2	D_1	D_0	CP	CP_0	CP_1	CP_2
0	0	0	0	0	1	×	0	1	↓	↓	↓	
0	0	1	0	1	0	0	1	0	↓	↓	↓	↓
0	1	0	0	1	1	×	1	1	↓	↓	↓	
0	1	1	1	0	0	1	0	0	↓	↓	↓	↓
1	0	0	1	1	1	×	1	1	↓	↓	↓	
1	0	1	0	0	0	0	0	0	↓	↓	↓	↓
1	1	0	1	0	1	×	0	1	↓	↓	↓	
1	1	1	1	1	0	1	1	0	↓	↓	↓	↓

（3） 卡诺图及驱动方程

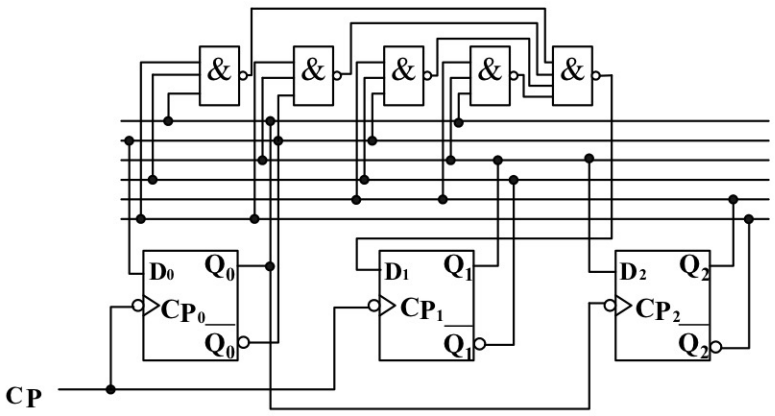
Q_0^n \ $Q_2^nQ_1^n$	00	01	11	10
0	×	×	×	×
1	0	1	1	0

Q_0^n \ $Q_2^nQ_1^n$	00	01	11	10
0	0	1	0	1
1	1	0	1	0

Q_0^n \ $Q_2^nQ_1^n$	00	01	11	10
0	1	1	1	1
1	0	0	0	0

$$D_2 = Q_1^n \qquad D_1 = \overline{\overline{Q}_2^n \overline{Q}_1^n Q_0^n \; \overline{Q}_2^n Q_1^n Q_0^n \; Q_2^n \overline{Q}_1^n \overline{Q}_0^n \; \overline{Q_2^n Q_1^n Q_0^n}} \qquad D_0 = \overline{Q}_0^n$$

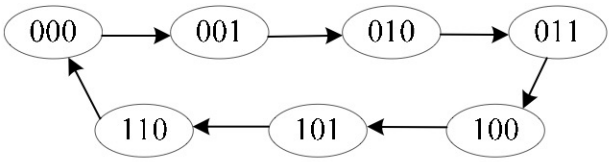
（4）电路图



第五章习题

5-1

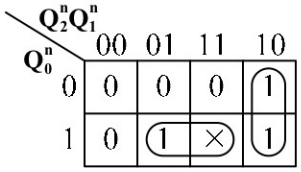
（1）时序图：



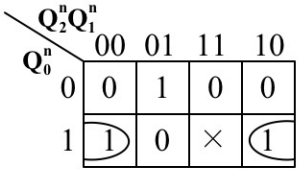
（2）状态转换和驱动真值表：

Q_2^n	Q_1^n	Q_0^n	Q_2^{n+1}	Q_1^{n+1}	Q_0^{n+1}	D_2	D_1	D_0
0	0	0	0	0	1	0	0	1
0	0	1	0	1	0	0	1	0
0	1	0	0	1	1	0	1	1
0	1	1	1	0	0	1	0	0
1	0	0	1	0	1	1	0	1
1	0	1	1	1	0	1	1	0
1	1	0	0	0	0	0	0	0
1	1	1	×	×	×	×	×	×

(3)卡诺图：



$$D_2 = Q_1^n Q_0^n + Q_2^n \overline{Q}_1^n \quad D_2 = \overline{\overline{Q_1^n Q_0^n} \cdot \overline{Q_2^n \overline{Q}_1^n}}$$

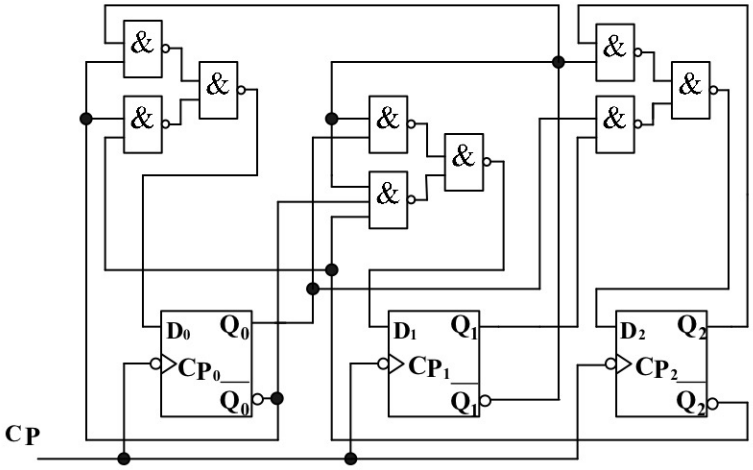


$$D_1 = \overline{Q}_1^n Q_0^n + \overline{Q}_2^n Q_1^n \overline{Q}_0^n \quad D_1 = \overline{\overline{\overline{Q}_1^n Q_0^n} \cdot \overline{\overline{Q}_2^n Q_1^n \overline{Q}_0^n}}$$

$Q_2^nQ_1^n$ / Q_0^n	00	01	11	10
0	1	1	0	1
1	0	0	×	0

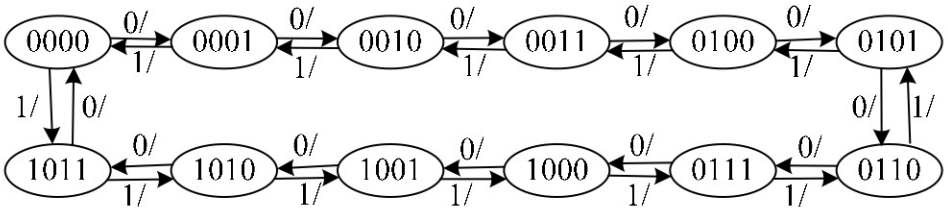
$D_0 = \overline{Q}_2^n\overline{Q}_0^n + \overline{Q}_1^n\overline{Q}_0^n$， $D_0 = \overline{\overline{\overline{Q}_2^n\overline{Q}_0^n} \cdot \overline{\overline{Q}_1^n\overline{Q}_0^n}}$

（4）电路图：



5-2

（1）时序图：

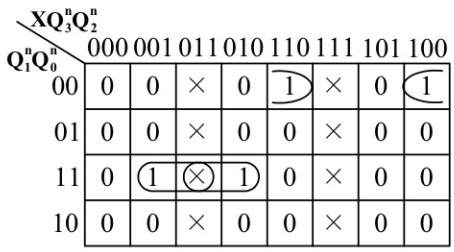


（2）状态转换和驱动真值表：

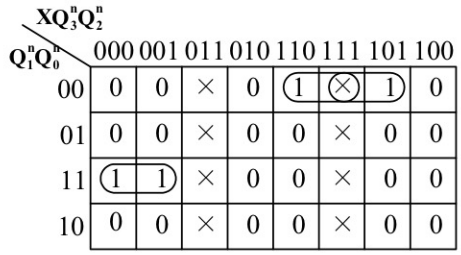
X	Q_3^n	Q_2^n	Q_1^n	Q_0^n	Q_3^{n+1}	Q_2^{n+1}	Q_1^{n+1}	Q_0^{n+1}	T_3	T_2	T_1	T_0
0	0	0	0	0	0	0	0	1	0	0	0	1
0	0	0	0	1	0	0	1	0	0	0	1	1
0	0	0	1	0	0	0	1	1	0	0	0	1
0	0	0	1	1	0	1	0	0	0	1	1	1
0	0	1	0	0	0	1	0	1	0	0	0	1
0	0	1	0	1	0	1	1	0	0	0	1	1
0	0	1	1	0	0	1	1	1	0	0	0	1
0	0	1	1	1	1	0	0	0	1	1	1	1
0	1	0	0	0	1	0	0	1	0	0	0	1
0	1	0	0	1	1	0	1	0	0	0	1	1

0	1	0	1	0	1	0	1	1	0	0	0	1
0	1	0	1	1	0	0	0	0	1	0	1	1
0	1	1	0	0	×	×	×	×	×	×	×	×
0	1	1	0	1	×	×	×	×	×	×	×	×
0	1	1	1	0	×	×	×	×	×	×	×	×
0	1	1	1	1	×	×	×	×	×	×	×	×
1	0	0	0	0	1	0	1	1	1	0	1	1
1	0	0	0	1	0	0	0	0	0	0	0	1
1	0	0	1	0	0	0	0	1	0	0	1	1
1	0	0	1	1	0	0	1	0	0	0	0	1
1	0	1	0	0	0	0	1	1	0	1	1	1
1	0	1	0	1	0	1	0	0	0	0	0	1
1	0	1	1	0	0	1	0	1	0	0	1	1
1	0	1	1	1	0	1	1	0	0	0	0	1
1	1	0	0	0	0	1	1	1	1	1	1	1
1	1	0	0	1	1	0	0	0	0	0	0	1
1	1	0	1	0	1	0	0	1	0	0	1	1
1	1	0	1	1	1	0	1	0	0	0	0	1
1	1	1	0	0	×	×	×	×	×	×	×	×
1	1	1	0	1	×	×	×	×	×	×	×	×
1	1	1	1	0	×	×	×	×	×	×	×	×
1	1	1	1	1	×	×	×	×	×	×	×	×

(3)卡诺图：



$$T_3 = \bar{X}Q_2^nQ_1^nQ_0^n + \bar{X}Q_3^nQ_1^nQ_0^n + X\bar{Q}_2^n\bar{Q}_1^n\bar{Q}_0^n$$



$$T_2 = \bar{X}\bar{Q}_3^nQ_1^nQ_0^n + XQ_3^n\bar{Q}_1^n\bar{Q}_0^n + XQ_2^n\bar{Q}_1^n\bar{Q}_0^n$$

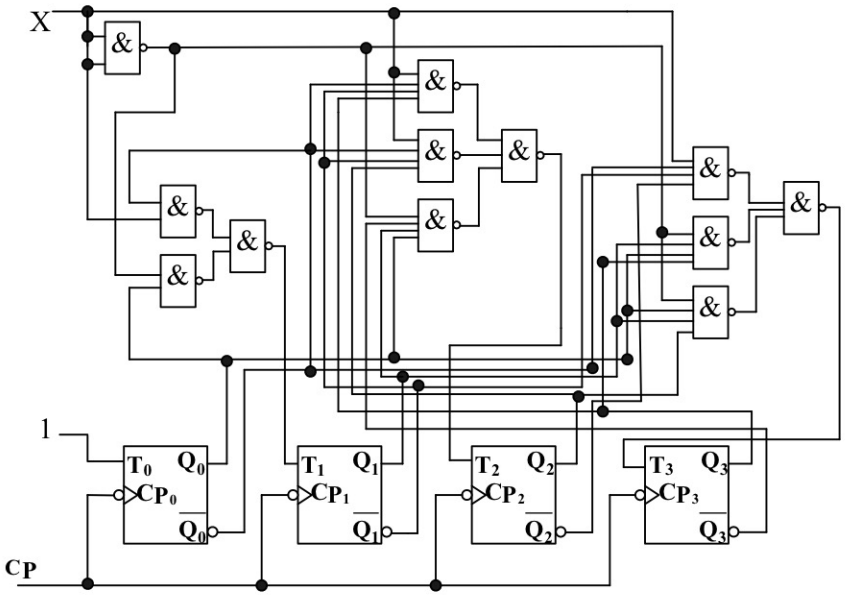
$XQ_3^nQ_2^n$ / $Q_1^nQ_0^n$	000	001	011	010	110	111	101	100
00	0	0	×	0	1	×	1	1
01	1	1	×	1	0	×	0	0
11	1	1	×	1	0	×	0	0
10	0	0	×	0	1	×	1	1

$$T_1 = \bar{X}Q_0^n + X\bar{Q}_0^n$$

$XQ_3^nQ_2^n$ / $Q_1^nQ_0^n$	000	001	011	010	110	111	101	100
00	1	1	×	1	1	×	1	1
01	1	1	×	1	1	×	1	1
11	1	1	×	1	1	×	1	1
10	1	1	×	1	1	×	1	1

$$T_0 = 1$$

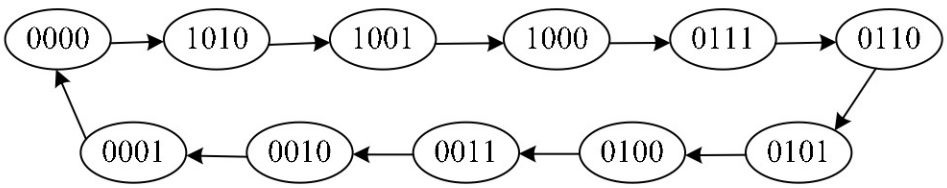
（4）电路图：



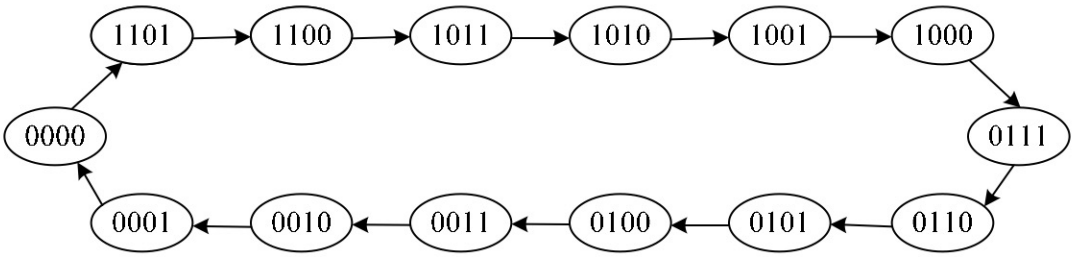
5-3

（1）状态图：X 为控制变量

X=0 时的状态图：



X=1 时的状态图：



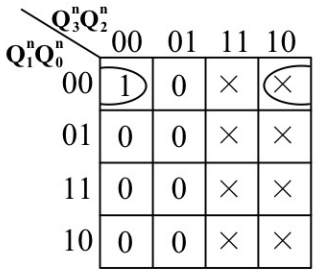
（2）状态转换及真值表：

X	Q_3^n	Q_2^n	Q_1^n	Q_0^n	Q_3^{n+1}	Q_2^{n+1}	Q_1^{n+1}	Q_0^{n+1}	J_3	K_3	J_2	K_2	J_1	K_1	J_0	K_0
0	0	0	0	0	1	0	1	0	1	×	0	×	1	×	0	×
0	0	0	0	1	0	0	0	0	0	×	0	×	0	×	×	1
0	0	0	1	0	0	0	0	1	0	×	0	×	×	1	1	×
0	0	0	1	1	0	0	1	0	0	×	0	×	×	0	×	1
0	0	1	0	0	0	0	1	1	0	×	×	1	1	×	1	×
0	0	1	0	1	0	1	0	0	0	×	×	0	0	×	×	1
0	0	1	1	0	0	1	0	1	0	×	×	0	×	1	1	×
0	0	1	1	1	0	1	1	0	0	×	×	0	×	0	×	1
0	1	0	0	0	0	1	1	1	×	1	1	×	1	×	1	×
0	1	0	0	1	1	0	0	0	×	0	0	×	×	0	×	1
0	1	0	1	0	1	0	0	1	×	0	0	×	×	1	1	×
0	1	0	1	1	×	×	×	×	×	×	×	×	×	×	×	×
0	1	1	0	0	×	×	×	×	×	×	×	×	×	×	×	×
0	1	1	0	1	×	×	×	×	×	×	×	×	×	×	×	×
0	1	1	1	0	×	×	×	×	×	×	×	×	×	×	×	×
0	1	1	1	1	×	×	×	×	×	×	×	×	×	×	×	×
1	0	0	0	0	1	1	1	0	1	×	1	×	1	×	0	×
1	0	0	0	1	0	0	0	0	0	×	0	×	0	×	×	1
1	0	0	1	0	0	0	0	1	0	×	0	×	×	1	1	×
1	0	0	1	1	0	0	1	0	0	×	0	×	×	0	×	1
1	0	1	0	0	0	0	1	1	0	×	×	1	1	×	1	×
1	0	1	0	1	0	1	0	0	0	×	×	0	0	×	×	1
1	0	1	1	0	0	1	0	1	0	×	×	0	×	1	1	×
1	0	1	1	1	0	1	1	0	0	×	×	0	×	0	×	1
1	1	0	0	0	0	1	1	1	×	1	1	×	1	×	1	×
1	1	0	0	1	1	0	0	0	×	0	×	0	0	×	×	1
1	1	0	1	0	1	0	0	1	×	0	×	0	×	1	1	×
1	1	0	1	1	1	0	1	0	×	0	×	0	×	0	×	1
1	1	1	0	0	1	0	1	1	×	0	×	1	1	×	1	×

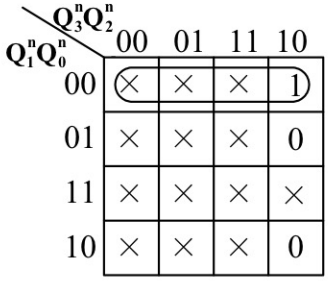
1	1	1	0	1	1	1	0	0	×	0	×	0	0	×	×	1
1	1	1	1	0	×	×	×	×	×	×	×	×	×	×	×	×
1	1	1	1	1	×	×	×	×	×	×	×	×	×	×	×	×

(3)卡诺图：

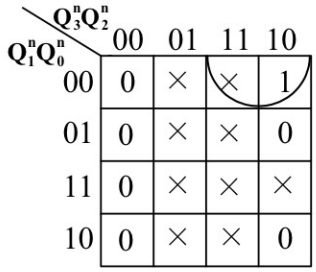
X=0 时：



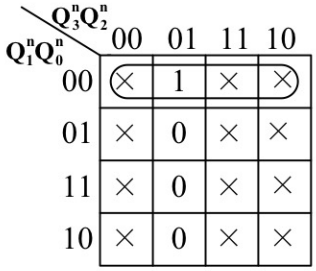
$$J_3 = \overline{Q}_2^n Q_1^n \overline{Q}_0^n$$



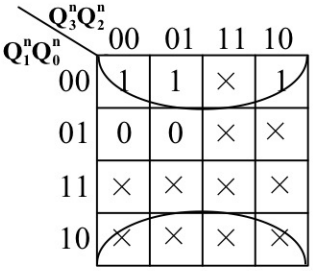
$$K_3 = \overline{Q}_1^n \overline{Q}_0^n$$



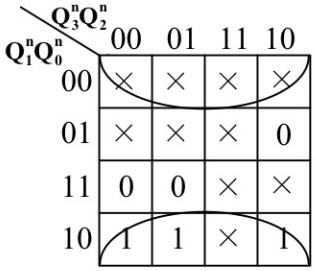
$$J_2 = Q_3^n \overline{Q}_1^n \overline{Q}_0^n$$



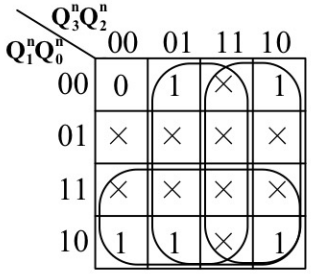
$$K_2 = \overline{Q}_1^n \overline{Q}_0^n$$



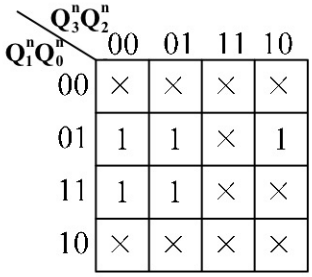
$$J_1 = \overline{Q}_0^n$$



$$K_1 = \overline{Q}_0^n$$

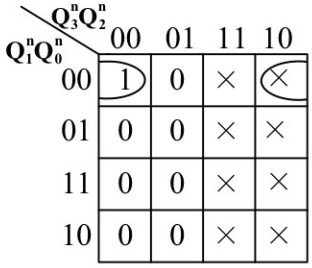


$$J_0 = Q_1^n + Q_2^n + Q_3^n$$

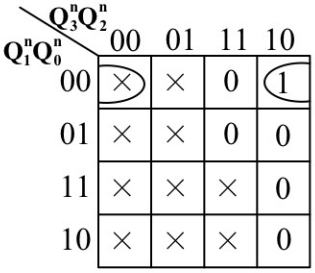


$$K_0 = 1$$

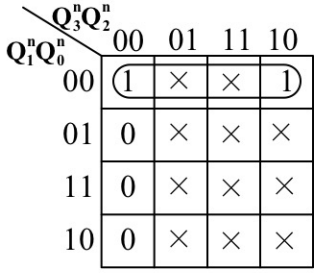
X=1 时：



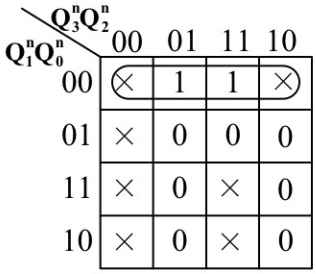
$$J_3 = \overline{Q}_2^n \overline{Q}_1^n \overline{Q}_0^n$$



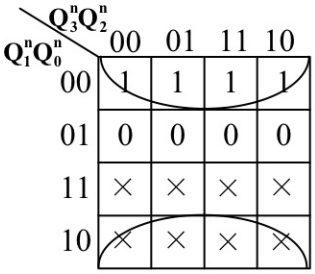
$$K_3 = \overline{Q}_2^n \overline{Q}_1^n \overline{Q}_0^n$$



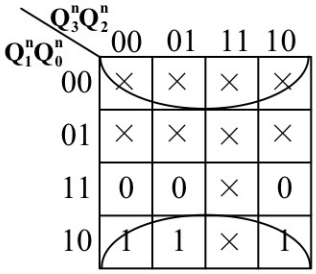
$$J_2 = \overline{Q}_1^n \overline{Q}_0^n$$



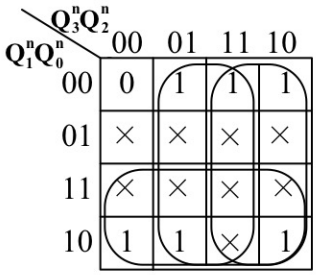
$$K_2 = \overline{Q}_1^n \overline{Q}_0^n$$



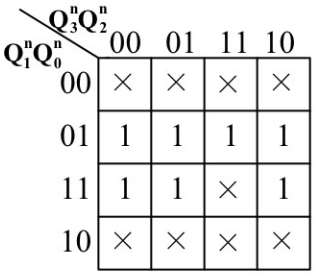
$$J_1 = \overline{Q}_0^n$$



$$K_1 = \overline{Q}_0^n$$



$$J_0 = Q_1^n + Q_2^n + Q_3^n$$



$$K_0 = 1$$

那么，有：

$$J_3 = \overline{Q}_2^n \overline{Q}_1^n \overline{Q}_0^n$$

$$K_3 = \overline{X}\overline{Q}_1^n \overline{Q}_0^n + X\overline{Q}_2^n \overline{Q}_1^n \overline{Q}_0^n = \overline{Q}_1^n \overline{Q}_0^n (\overline{X} + X\overline{Q}_2^n) = \overline{X}\overline{Q}_1^n \overline{Q}_0^n + \overline{Q}_2^n \overline{Q}_1^n \overline{Q}_0^n$$

$$J_2 = \overline{X}Q_3^n \overline{Q}_1^n \overline{Q}_0^n + X\overline{Q}_1^n \overline{Q}_0^n = Q_3^n \overline{Q}_1^n \overline{Q}_0^n + X\overline{Q}_1^n \overline{Q}_0^n$$

$$K_2 = \overline{Q}_1^n \overline{Q}_0^n$$

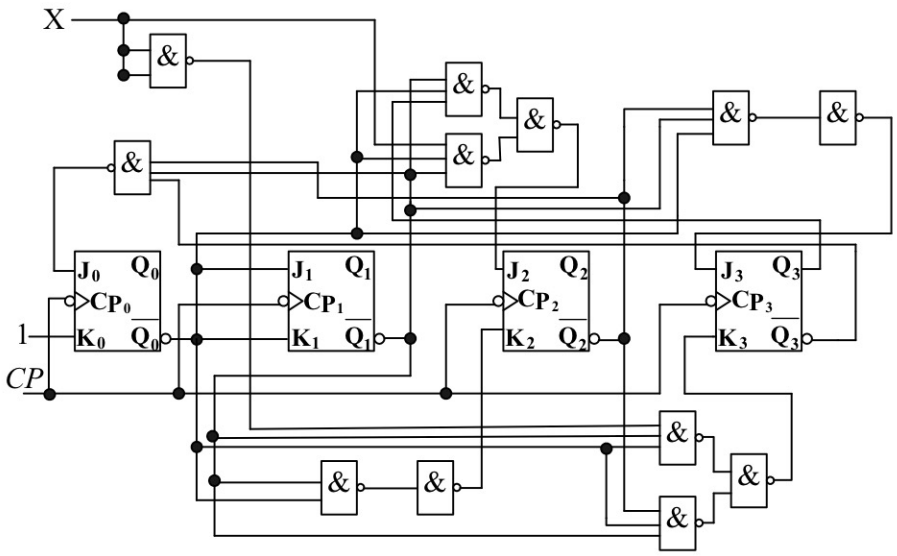
$$J_1 = \overline{Q}_0^n$$

$$K_1 = \overline{Q}_0^n$$

$$J_0 = Q_1^n + Q_2^n + Q_3^n = \overline{\overline{Q}_1^n \overline{Q}_2^n \overline{Q}_3^n}$$

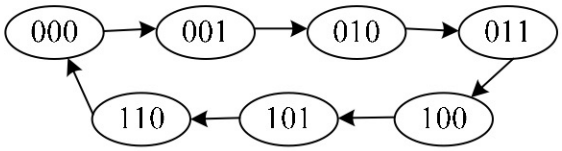
$K_0 = 1$

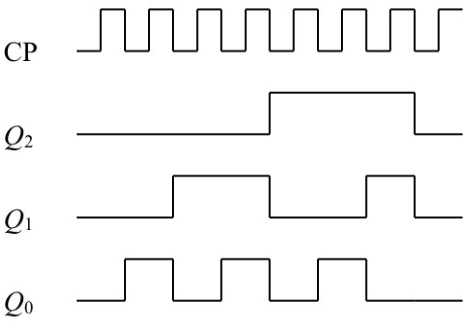
（4）电路图：



5-4

（1）时序图：





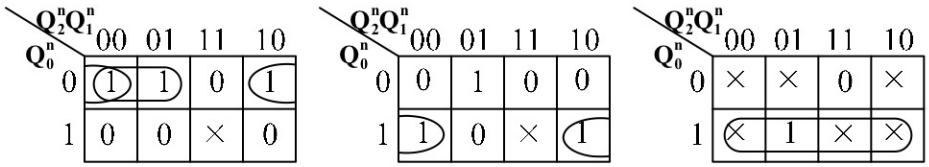
可见：$CP_0 = CP\downarrow$，$CP_1 = CP\downarrow$，$CP_2 = Q_1\downarrow$

(2)状态转换图及真值表：

Q_2^n	Q_1^n	Q_0^n	Q_2^{n+1}	Q_1^{n+1}	Q_0^{n+1}	D_2	D_1	D_0	CP	CP_0	CP_1	CP_2
0	0	0	0	0	1	×	0	1	↓	↓	↓	
0	0	1	0	1	0	×	1	0	↓	↓	↓	
0	1	0	0	1	1	×	1	1	↓	↓	↓	
0	1	1	1	0	0	1	0	0	↓	↓	↓	↓

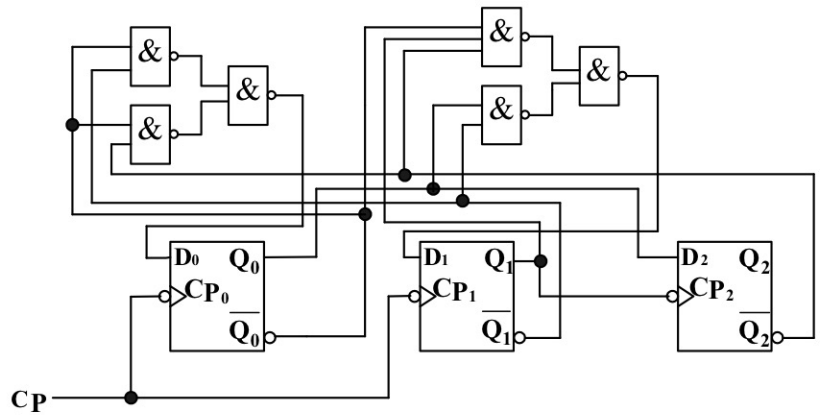
1	0	0	1	0	1	×	0	1	↓	↓	↓	
1	0	1	1	1	0	×	1	0	↓	↓	↓	
1	1	0	0	0	0	0	0	0	↓	↓	↓	↓
1	1	1	×	×	×	×	×	×	↓	↓	↓	

(3)卡诺图：



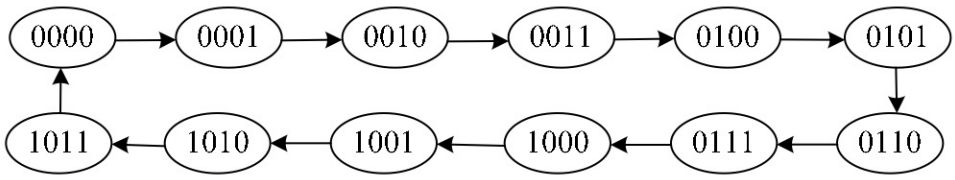
$$D_0 = \bar{Q}_2^n\bar{Q}_0^n + \bar{Q}_1^n\bar{Q}_0^n \qquad D_1 = \bar{Q}_2^nQ_1^n\bar{Q}_0^n + \bar{Q}_1^nQ_0^n \qquad D_2 = Q_0^n$$

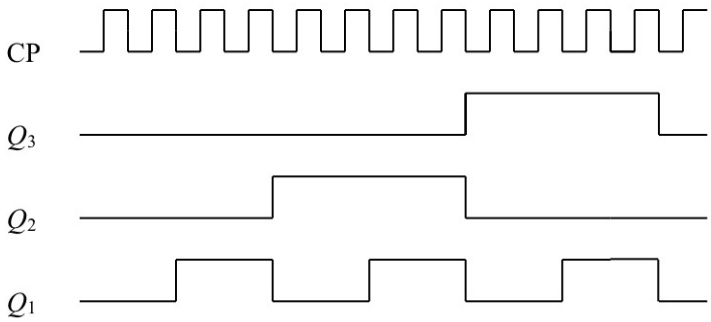
（4）电路图：

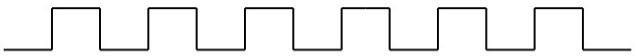


5-5

（1）时序图：





Q_0 

可见：$CP_0 = CP\downarrow$，$CP_1 = Q_0\downarrow$，$CP_2 = Q_1\downarrow$，$CP_3 = Q_1\downarrow$

（2）状态转换及真值表：

Q_3^n	Q_2^n	Q_1^n	Q_0^n	Q_3^{n+1}	Q_2^{n+1}	Q_1^{n+1}	Q_0^{n+1}	T_3	T_2	T_1	T_0	CP	CP_0	CP_1	CP_2	CP_3
0	0	0	0	0	0	0	1	×	×	×	1	↓	↓			
0	0	0	1	0	0	1	0	×	×	1	1	↓	↓	↓		
0	0	1	0	0	0	1	1	×	×	×	1	↓	↓			
0	0	1	1	0	1	0	0	0	1	1	1	↓	↓	↓	↓	↓
0	1	0	0	0	1	0	1	×	×	×	1	↓	↓			
0	1	0	1	0	1	1	0	×	×	1	1	↓	↓	↓		
0	1	1	0	0	1	1	1	×	×	×	1	↓	↓			
0	1	1	1	1	0	0	0	1	1	1	1	↓	↓	↓	↓	↓
1	0	0	0	1	0	0	1	×	×	×	1	↓	↓			
1	0	0	1	1	0	1	0	×	×	1	1	↓	↓	↓		
1	0	1	0	1	0	1	1	×	×	×	1	↓	↓			
1	0	1	1	0	0	0	0	1	0	1	1	↓	↓	↓	↓	↓
1	1	0	0	×	×	×	×	×	×	×	×	↓	↓			
1	1	0	1	×	×	×	×	×	×	×	×	↓	↓			
1	1	1	0	×	×	×	×	×	×	×	×	↓	↓			
1	1	1	1	×	×	×	×	×	×	×	×	↓	↓			

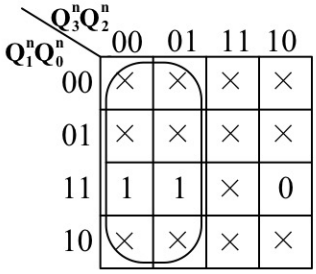
(3)卡诺图：

$Q_1^nQ_0^n$ \ $Q_3^nQ_2^n$	00	01	11	10
00	1	1	×	1
01	1	1	×	1
11	1	1	×	1
10	1	1	×	1

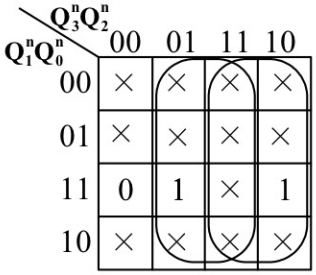
$T_0 = 1$

$Q_1^nQ_0^n$ \ $Q_3^nQ_2^n$	00	01	11	10
00	×	×	×	×
01	1	1	×	1
11	1	1	×	1
10	×	×	×	×

$T_1 = 1$

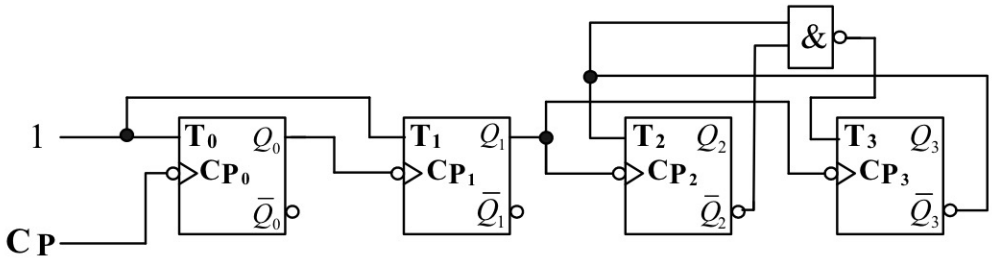


$T_2 = \overline{Q}_3^n$



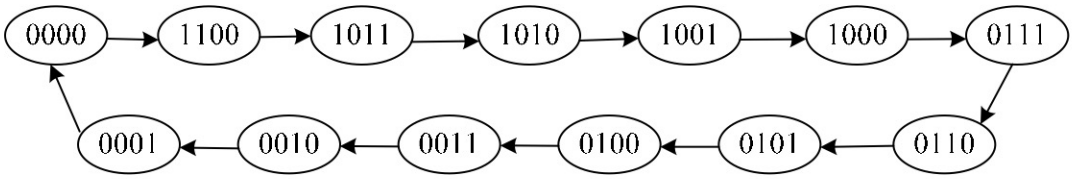
$T_3 = \overline{\overline{Q}_2^n \overline{Q}_3^n}$

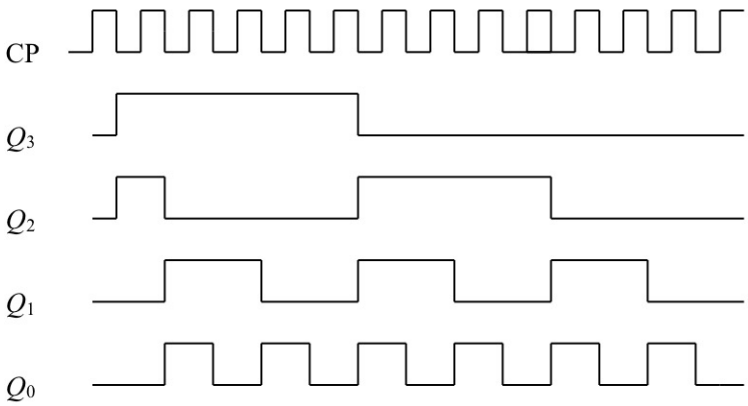
（4）电路图：



5-6

（1）时序图：



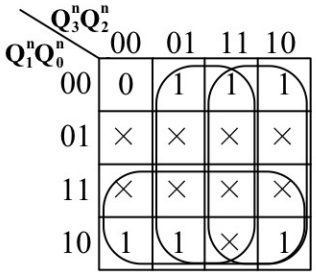


可见：$CP_0 = CP\downarrow$，$CP_1 = \overline{Q}_0\downarrow$，$CP_2 = CP\downarrow$，$CP_3 = \overline{Q}_2\downarrow$

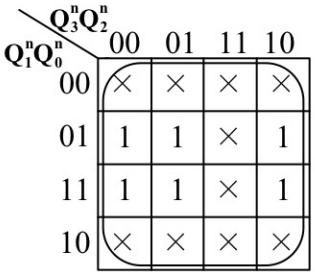
（2）状态转换真值表：

Q_3^n			Q_2^n	Q_3^{n+1}	Q_2^{n+1}		Q_1^{n+1}	J_3	K_3	J_2	K_2	J_1	K_1	J_0	K_0
Q_1^n	Q_0^n			Q_0^{n+1}											
0	0	0	0	1	1	0	0	1	×	1	×	×	×	0	×
0	0	0	1	0	0	0	0	×	×	0	×	×	×	×	1
0	0	1	0	0	0	0	1	×	×	0	×	×	1	1	×
0	0	1	1	0	0	1	0	×	×	0	×	×	×	×	1
0	1	0	0	0	0	1	1	×	×	×	1	1	×	1	×
0	1	0	1	0	1	0	0	×	×	×	0	×	×	×	1
0	1	1	0	0	1	0	1	×	×	×	0	×	1	1	×
0	1	1	1	0	1	1	0	×	×	×	0	×	×	×	1
1	0	0	0	0	1	1	1	×	1	1	×	1	×	1	×
1	0	0	1	1	0	0	0	×	×	0	×	×	×	×	1
1	0	1	0	1	0	0	1	×	×	0	×	×	1	1	×
1	0	1	1	1	0	1	0	×	×	0	×	×	×	×	1
1	1	0	0	1	0	1	1	×	×	×	1	1	×	1	×
1	1	0	1	×	×	×	×	×	×	×	×	×	×	×	×
1	1	1	0	×	×	×	×	×	×	×	×	×	×	×	×
1	1	1	1	×	×	×	×	×	×	×	×	×	×	×	×

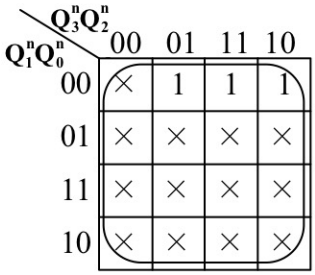
(3) 卡诺图：



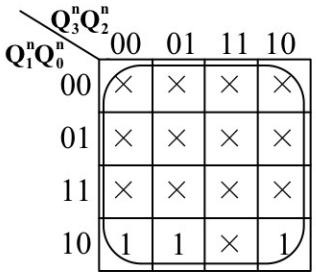
$$J_0 = \overline{\overline{Q}_1^n \overline{Q}_2^n \overline{Q}_3^n}$$



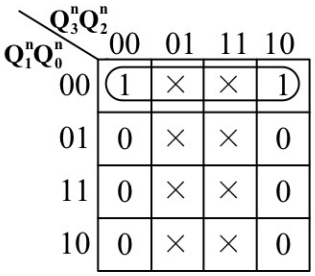
$$K_0 = 1$$



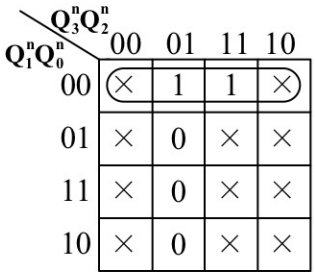
$$J_1 = 1$$



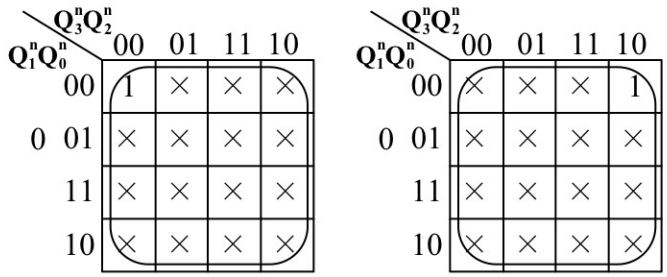
$$K_1 = 1$$



$$J_2 = \overline{Q}_1^n \overline{Q}_0^n$$

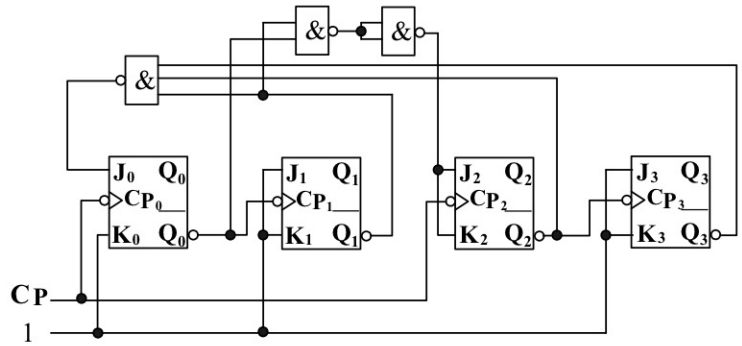


$$K_2 = \overline{Q}_1^n \overline{Q}_0^n$$



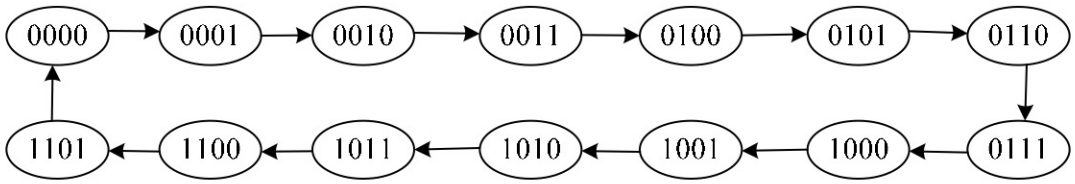
$J_3 = 1$ $K_3 = 1$

(4)电路图：



5-7

状态图：

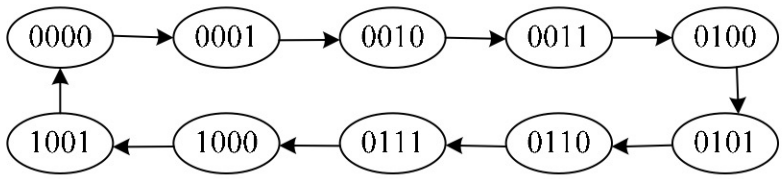


因此，计数模值为 14.

5-8

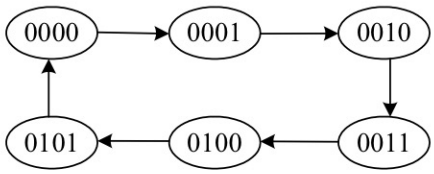
$$\overline{LD} = \overline{\overline{C}Q_3^nQ_0^n + CQ_2^nQ_0^n}$$

C=0 代入上式，则有 $\overline{LD} = \overline{Q_3^nQ_0^n}$， 状态图为：



因此，计数模值为 10.

同理，C=1 代入，$\overline{LD}=\overline{Q_2^nQ_0^n}$，状态图：



因此，计数模值为 6.

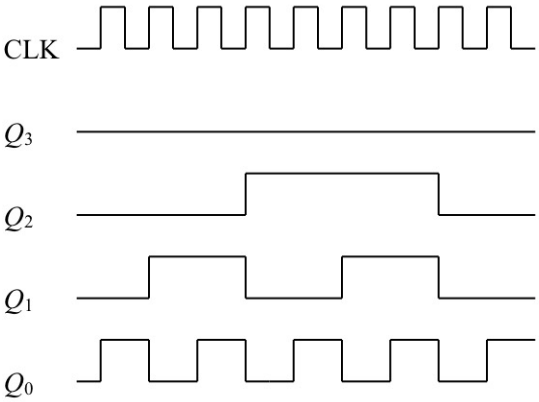
5-9

计数模值为 29.

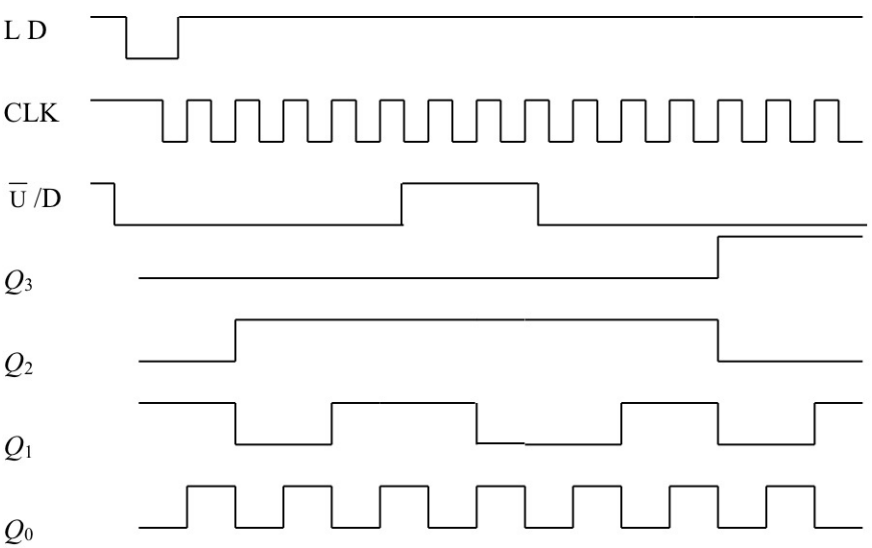
5-10

计数模值为 39.

5-11

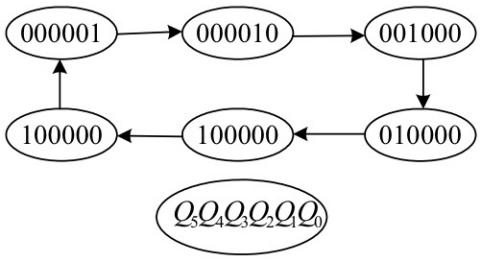


5-12

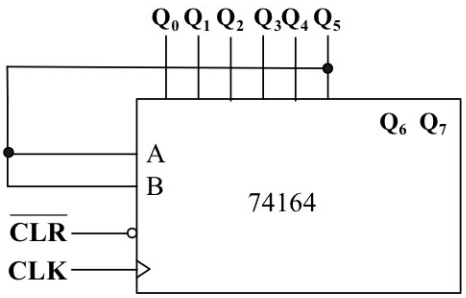


5-13

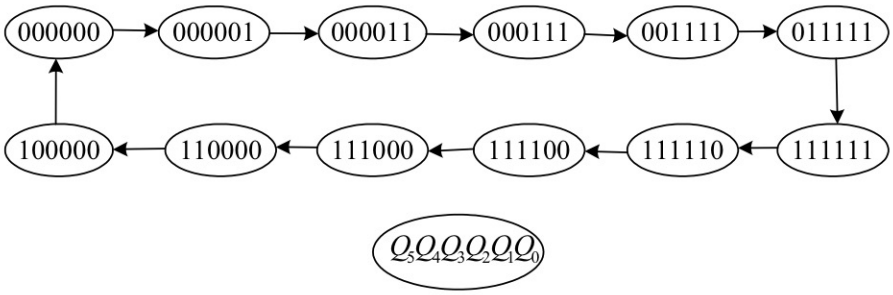
（1） $A=B=Q_5^n$，状态图为：



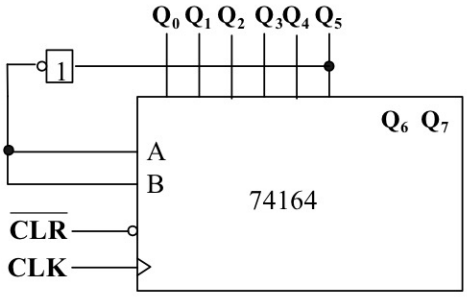
电路图为：



（2） $A=B=\overline{Q}_5^n$，状态图为：

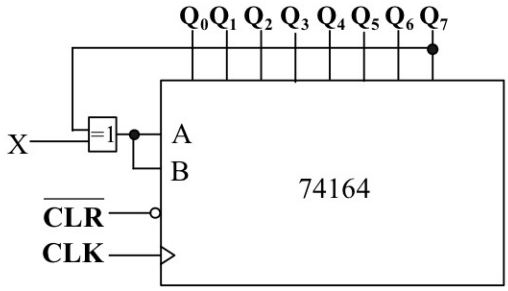


电路图为：



5-14

X 为控制信号， $A = B = X\overline{Q}_7^n + \overline{X}Q_7^n$



5-15

$A = \overline{Q}_4 C$ ， $B = D$ ， $Q_0 = AB = \overline{Q}_4 CD$

